

PQ-DID: A Generic Framework for Post-Quantum Decentralised Identity with Privacy-Preserving Credential Authentication and Revocation

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Abstract—Decentralised identity systems enable user-centric credential management and privacy-preserving authentication without relying on centralised identity providers. However, existing decentralised identity deployments are predominantly built on classical cryptographic assumptions and therefore do not provide long-term security against quantum adversaries. Simply replacing classical signatures with post-quantum signatures is insufficient, since a post-quantum decentralised identity system must also support reusable credentials, selective disclosure, unlinkable multi-verifier authentication, and credential revocation within a unified lifecycle.

This paper presents PQ-DID, a generic post-quantum decentralised identity framework that formalises credential issuance, privacy-preserving presentation, verification, and revocation while remaining compatible with W3C DID and Verifiable Credentials workflows. To realise the authentication layer of this framework, we introduce PQ-DAA, a post-quantum decentralised anonymous authentication primitive that enables holders to prove possession of valid issuer-certified credentials without exposing persistent holder-side public verification keys. We formalise the syntax and security requirements of PQ-DID and PQ-DAA, provide game-based security models, and present a lattice-based instantiation using ML-DSA, lattice commitments, zero-knowledge proofs, and accumulator-based revocation.

We establish the security of the concrete construction in the Quantum Random Oracle Model under Module-SIS, Module-LWE, ML-DSA EUF-CMA security, accumulator soundness, and the stated security assumptions of the underlying proof system. We further implement a proof-of-concept prototype supporting credential issuance, anonymous authentication, selective disclosure, presentation verification, and revocation checking. The experimental results demonstrate the feasibility of privacy-preserving decentralised identity in the post-quantum setting.

Index Terms—post-quantum, decentralised identity (DID), authentication

I. INTRODUCTION

DECENTRALISED Identity (DID) systems have emerged as a promising alternative to traditional centralised identity infrastructures. By decoupling identity management from centralised authorities and leveraging a decentralised architecture, DID systems empower users to maintain control over their sensitive identity information while disclosing only the minimum personal information required, significantly enhancing user privacy. Typically, current DID are formalised through

the W3C Decentralised Identifiers (DID) specification and the Verifiable Credentials (VC) data model [1], [2].

Given the great potential to improve personal privacy, pioneering countries and regions have begun adopting DID in their national infrastructure. For example, the European Union is piloting the EU Digital Identity Wallet, which enables citizens to access government services, make payments, and authenticate online in compliance with the eIDAS 2.0 regulation, Bhutan migrated its National Digital Identity (NDI) system to the Ethereum mainnet in October 2025, providing citizens with secure digital identities, while South Korea launched a mobile resident registration card in 2025, allowing citizens to store their digital identity securely in a blockchain-based mobile application.

However, despite the advances and the large potential for real-world deployment, most existing decentralised identity systems remain vulnerable to quantum adversaries because they rely on classical cryptographic primitives that are not resistant to quantum attacks [1]–[4]. As summarised in Table I, current DID systems provide mature identity management and privacy-preserving functionalities, but are built upon classical cryptographic assumptions. Although several recent works have attempted to address this limitation by replacing classical signatures with post-quantum alternatives [5]–[8], Table ?? shows that these efforts primarily focus on individual cryptographic components or authentication mechanisms. Consequently, primitive replacement alone does not provide a complete post-quantum decentralised identity architecture.

Post-quantum authentication research has produced lattice-based identification protocols [9], anonymous authentication schemes [10], [11], and privacy-preserving credentials [7], [12]. These constructions provide important cryptographic building blocks but remain largely protocol-centric: they do not jointly capture DID resolution, reusable credential issuance, selective disclosure, unlinkable multi-verifier presentation, and revocation. Hence, replacing individual classical primitives is insufficient to obtain a complete post-quantum DID architecture.

To address this gap, we propose PQ-DID, a generic post-quantum decentralised identity framework covering identifier management, credential issuance, privacy-preserving presentation, verification, and revocation, as illustrated in Figure I. PQ-DID separates identity management from authentication: DID documents provide identifier resolution and public metadata, while holders authenticate by proving possession of issuer-certified credentials. We realise this authentication

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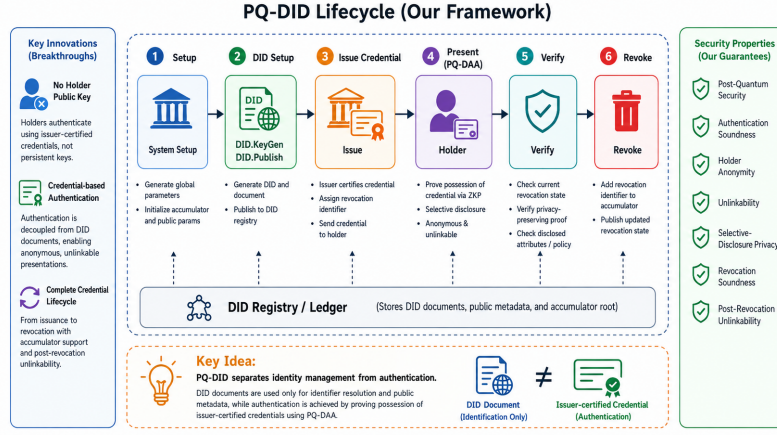


Fig. 1. Overview of the proposed PQ-DID framework.

layer through PQ-DAA and provide a concrete lattice-based instantiation and proof-of-concept prototype.

From the technical perspective, to instantiate the authentication layer of PQ-DID, we introduce post-quantum decentralised anonymous authentication (PQ-DAA). PQ-DAA is not intended as a standalone identity system, rather, it serves as the cryptographic layer that enables anonymous credential presentation, selective disclosure, unlinkability, and revocation checking inside the PQ-DID architecture. We further present a concrete lattice-based instantiation and a proof-of-concept implementation to demonstrate that the proposed architecture can be realised with practical post-quantum primitives.

A. Our Contributions

This paper makes the following contributions.

- **Theoretical contributions.** We propose PQ-DID, a generic post-quantum decentralised identity framework that captures the complete credential lifecycle, including identifier management, credential issuance, privacy-preserving presentation, verification, and revocation. The framework is primitive-agnostic and remains compatible with W3C DID and Verifiable Credentials workflows.
- We introduce PQ-DAA, a post-quantum decentralised anonymous authentication primitive that serves as the authentication layer of PQ-DID. PQ-DAA supports reusable credentials, selective disclosure, unlinkable multi-verifier authentication, and revocation-aware verification.
- We formalise the syntax and game-based security requirements of both PQ-DID and PQ-DAA, including authentication soundness, anonymity, unlinkability, selective disclosure privacy, revocation soundness, and forward security.
- We present a concrete lattice-based instantiation using ML-DSA, lattice commitments, zero-knowledge proofs, and a dynamic accumulator. We establish security in the QROM under Module-SIS, Module-LWE, ML-DSA EUF-CMA security, accumulator soundness, and the stated proof-system assumptions.
- **Practical contribution.** We implement a proof-of-concept prototype and evaluate its computation, storage,

and communication costs, demonstrating the feasibility of privacy-preserving decentralised identity in the post-quantum setting.

II. RELATED WORK

We review decentralised identity, post-quantum self-sovereign identity (SSI), anonymous credentials and revocation, focusing on their relationship to PQ-DID's credential lifecycle and privacy model.

A. Decentralised Identity and Privacy-Preserving Credentials

The W3C DID and Verifiable Credentials specifications define identifier and credential data models; security depends on the selected cryptographic mechanisms [13], [14]. Anon-Creds v1.0 combines Camenisch–Lyssanskaya signatures over RSA groups with pairing-based revocation, while the BBS cryptosuite uses pairing-based signatures. Both support selective disclosure and unlinkable presentations under classical assumptions [15], [16].

DISC supports multi-authority credential aggregation, unlinkable authentication and batch verification [17]. DP-DID addresses dynamic and proactive identity management [18], while FRAC expresses access-control rules and addresses credential theft and revoked-credential reuse [19]. PQ-DID studies quantum-resistant credential presentation and revocation-related privacy. Aggregation, proactive key protection and a general-purpose access-control language are outside its scope.

B. Post-Quantum Decentralised Identity and SSI

Pino et al. studied hybrid post-quantum and classical credentials and presentations for IoT [20]. Solavagione and Vesco implemented pure post-quantum and hybrid SSI in IOTA Identity, covering DID, VC and VP workflows [21]. Zeydan et al. developed post-quantum blockchain-based identity management for resource sharing in open radio access networks [22].

Presentation privacy has also been addressed directly. Margaria et al. adapted Bootle et al.'s lattice anonymous-credential framework to SSI and implemented selective disclosure in

C++ [23], [24]. Boi et al. investigated ZK-STARKs for post-quantum SSI and attribute-based authentication on edge devices [25]. The W3C Quantum-Resistant Cryptosuites draft specifies Data Integrity mechanisms based on post-quantum signatures [26]. These works cover both signature migration and privacy-preserving presentation, providing direct comparators for PQ-DID.

C. Post-Quantum Anonymous Credentials and Attestation

Bootle et al.'s BLNS framework supports selective disclosure and unlinkable presentations under specialised lattice assumptions, with privacy against coalitions including malicious issuers, holders and verifiers [24]. Argo et al. instantiated anonymous credentials with explicit proof encodings and a C implementation [27]. Both works also discuss revocation extensions and provide comparators for PQ-DAA's credential-proof layer.

Chen et al. constructed lattice-based direct anonymous attestation using collaborative, segregated non-interactive zero knowledge [28]. A trusted platform module and its host jointly generate attestations, with basename-controlled linkability and key-based revocation. Their hardware trust and collaborative proving architecture differ from PQ-DID's issuer-holder-verifier model.

D. Revocation and Lifecycle Privacy

Margaria et al. use validity epochs and issuer-provided credential updates [23]. PQ-DID proves that a hidden, issuer-certified identifier corresponds to an unrevoked leaf in a Merkle revocation tree; public updates support local witness maintenance. The mechanisms differ in status representation and update procedures, whose costs require empirical comparison. The IRAC preprint combines issuer hiding, flexible credentials and decentralised revocation with a classical implementation [29]; PQ-DID retains a public issuer.

PQ-DID binds selective disclosure and non-revocation to the same credential and presentation context. Its post-revocation unlinkability model includes historical presentations and subsequent public revocation information. The baseline permits colluding verifiers but assumes honest issuers and revocation managers whose private records remain undisclosed. Challenge holders remain uncorrupted throughout; issuer, schema, policy, epoch and disclosed attributes are public. These restrictions distinguish its guarantees from malicious-issuer privacy and security after holder compromise.

Table I summarises the comparisons. The reference construction in Section VII uses ML-DSA certification, hash-based holder binding, a Merkle revocation tree and an MPC-in-the-head argument for the complete authentication relation. Section VIII analyses its security under the stated component, quantum random-oracle model (QROM) and service assumptions.

III. BACKGROUND AND BUILDING BLOCKS

A. Architecture and Object Mapping

DID Core specifies identifiers, documents and method operations; the Verifiable Credentials (VC) data model specifies

credentials and presentations [13], [14]. PQ-DID follows the issuer-holder-verifier workflow: publication and resolution manage DID records, issuance produces a private credential record vc , presentation produces disclosed claims and a PQ-DAA proof vp , and revocation updates credential status through rs_e . This mapping is architectural: the protocol encodings alone do not establish VC/VP data-model conformance or standardisation of the reference DID method. Conforming representations require the corresponding data-model properties and proof/status extensions.

The reference method retains controller public keys in signed method records outside the minimal DID document. Credential authentication uses a separate holder secret x_H . Although a holder DID may be publicly registered, presentations neither disclose nor resolve it unless the holder-approved policy requires disclosure.

B. Roles and Trust Assumptions

The issuer validates evidence and approves the exact attribute vector, including the issuance-time DID and version reference (did, v_D) . The holder proves knowledge of its binding secret. The issuer checks controller authorisation and certifies the approved vector, holder binding and allocated revocation identifier. Issuance uses an authenticated confidential channel; the holder retains its secret, credential and revocation witness locally. Certification establishes issuer approval, the validity of external evidence depends on the issuer.

Application configuration fixes accepted issuer/key/schema instances, revocation namespaces and trusted authority, registry and audience-request keys. Resolution validates method state and controller authorisation, issuer trust remains an application decision. The revocation manager processes authorised requests and publishes authenticated states and updates, from which holders update witnesses locally.

The model assumes correctly generated parameters, consistent registry operations and an authenticated current-state service. The verifier's final ordered state read fixes the verification epoch, unavailable or inconsistent state causes rejection. Authenticity of an old signed state does not establish freshness. Holders approve authenticated requests, verifiers enforce the expected audience, session, policy and expiry, with trusted time and atomic challenge consumption. These service assumptions remain explicit in the lifecycle proofs.

C. Adversaries and Public Information

Adversaries are quantum polynomial-time (QPT), with classical protocol queries and quantum access to the proof random oracle. Enrolments are sequential, presentations may be repeated across colluding verifiers under changing supported policies. Corrupted holders may collude with verifiers. Setup, the targeted issuer, revocation manager and current-state service remain honest, and DID-method integrity is assumed. Privacy protects holders uncorrupted throughout the experiment, including after revocation, and excludes authority-verifier collusion and disclosure of private authority records. Acceptance attacks target an honest verifier.

TABLE I
RELATED WORK AND SCOPE OF PQ-DID

Work	Focus and relationship to PQ-DID
AnonCreds and BBS [15], [16]	Classical selective disclosure and unlinkable presentations; PQ-DID targets these functions under post-quantum assumptions.
DISC [17]	Multi-authority aggregation, unlinkable authentication and batch verification; aggregation and batching are outside PQ-DID's scope.
DP-DID [18]	Dynamic and proactive identity management; PQ-DID concerns credential presentation and revocation-related privacy.
FRAC [19]	Declarative access control and credential-misuse protection; PQ-DID focuses on quantum security and credential-lifecycle privacy.
Pino et al.; Solavagione and Vesco [20], [21]	Hybrid and pure post-quantum DID/VC/VP workflows; PQ-DID uses proofs of hidden credential witnesses.
Margaria et al. [23]	Lattice anonymous credentials for SSI, selective disclosure and epoch updates; PQ-DID uses a Merkle tree with hidden-identifier non-revocation proofs.
Boi et al. [25]	STARK-based SSI and attribute authentication on edge devices; an alternative proving architecture.
Bootle et al. [24]	Lattice anonymous credentials with malicious-issuer privacy; PQ-DID's baseline assumes honest issuers.
Argo et al. [27]	Concrete anonymous credentials with explicit proof encodings and implementation; a close comparator for the credential-proof layer.
Chen et al. [28]	TPM-host attestation with basenamespace-controlled linkability and key-based revocation; a different trust and proving architecture.
IRAC [29]	Issuer-hiding credentials and decentralised revocation with a classical instantiation; PQ-DID keeps the issuer public.
PQ-DID (this work)	Reference construction using ML-DSA certification, hash-based holder binding, a Merkle revocation tree and an MPC-in-the-head argument for context-bound authentication, under stated component, QROM and service assumptions.

Entries compare scope, not measured performance or guarantees under identical adversarial models. Omitted capabilities are not assumed absent; FRAC and Boi are limited to their reported scope. PQ-DID links issuer certification, holder-binding consistency and non-revocation for the same credential. Its specification and security analysis are distinct from implementation validation.

Issuers observe approved attributes, DID references, certified bindings and revocation identifiers, managers observe registrations and revocation requests. Public observations include method records, resolution queries, instance keys and schema, namespace, session context, policy, epoch, revocation root, disclosed positions and values, and complete proof bytes and lengths. Revocation updates additionally reveal the revoked identifier and its old sibling path. Section VI distinguishes authorised leakage from proof data and tests the joint view, including later revocation publications, under matching cumulative disclosures. Network addresses, traffic timing and availability are outside the credential-level model.

D. Cryptographic Interfaces

Signatures. KeyGen, Sign and Verify require correctness and post-quantum EUF-CMA security with classical signing queries. Section VII uses ML-DSA-65 with separate signing contexts for credentials and lifecycle messages [30], its bounded-operation losses are accounted for in Section VIII.

Holder binding and certification. BindRep recovers the certified representation B , BindOpen checks its secret/message opening, and CredValid additionally checks issuer certification and identifier consistency. The selected B contains a SHA3-384 holder-secret digest and the canonical approved attribute vector. Binding requires quantum collision

resistance; secrecy of the uniformly sampled secret given its digest additionally requires quantum preimage resistance. Attributes are visible to the issuer, presentation privacy is supplied by the proof.

Revocation. The status mechanism supplies state updates, witness generation, NRVerify and UpdateWit, requiring correctness and adaptive non-membership soundness. The selected mechanism is a SHA3-384 Merkle tree of revoked bits: an unrevoked credential has a path authenticating a zero leaf. State authentication and freshness are enforced by the lifecycle interfaces.

Arguments of knowledge. Prove and Check operate on the complete relation $\mathcal{R}_{\text{auth}}$ of Section V-B, jointly checking valid encodings, holder binding, issuer certification, authorised disclosure and non-revocation for the same certified identifier, with the presentation context bound to the statement. The selected Boolean MPC-in-the-head protocol uses raw random tapes, salted view commitments and Fiat-Shamir challenges [31]. Its requirements are completeness, adaptive quantum zero knowledge and history-preserving complete-witness extraction. Sections VII and VIII specify the circuits and derive the resource-dependent QROM bounds [32], [33]. The baseline supports disclosure and public checks on disclosed values, hidden-attribute predicates require an additional relation.

IV. GENERIC POST-QUANTUM DECENTRALISED IDENTITY FRAMEWORK

PQ-DID specifies a credential lifecycle comprising identifier management, issuance, privacy-preserving presentation, verification and revocation. The DID method supplies authenticated identifier management, while PQ-DAA supplies credential authentication. This section defines their interfaces and composition, Section VII specifies the construction against these interfaces.

A. Protocol State and Presentation Context

We adopt the roles and trust assumptions of Section III. Each instance fixes one issuer verification-key version, one credential schema and one revocation namespace, including its authorised state-verification key. An application may accept several such instances, but cross-instance credential aggregation and authority-key rotation are outside the baseline model. The reference ref_I identifies the issuer, key version and immutable schema encoding; a schema reference cannot subsequently be reassigned to a different encoding.

Public system parameters params contain the cryptographic parameters pp and the DID-method configuration. Holder state is $st_H = (x_H, st_{ctl})$, where x_H is the credential-binding secret and st_{ctl} supports method-specific controller authorisation. The domain and sampling distribution of x_H are specified by pp . Any credential-specific opening material is retained in the holder-local record vc , separately from its epoch-dependent revocation witness, that material may be empty in a particular instantiation. The supported message domains, canonical encodings, policy family and capacity limits are also fixed by pp .

Let $\mathcal{R}_e$ be the revoked set in namespace ns at epoch e . Its authenticated state and reference are

$$rs_e = (ns, e, A_e, \tau_e), \quad \text{ref}_e = (ns, e, A_e),$$

where A_e represents $\mathcal{R}_e$ and τ_e authenticates the namespace, epoch and state value. Write $\mu = (\text{ref}_I, ns)$ for the certified public metadata of the instance.

An honest verifier issues an authenticated request containing

$$ctx = (\text{tag}, \text{aud}, \text{sid}, n, \text{policy}, \\ \text{ref}_I, \text{ref}_e, t_{\text{exp}}).$$

The fields identify the protocol version, intended audience, application session, nonce, disclosure policy, issuer reference, revocation state and expiry time. The verifier samples n uniformly from $\{0, 1\}^{2\lambda}$, resampling a nonce already used by that audience. It records the complete context as pending in trusted verifier state st_V . Replicas sharing an audience maintain consistent nonce and session state. The holder authenticates the request, checks the audience and approves the policy before responding. The baseline policies are defined in Section V.

The presentation proof binds the complete context and all other public statement fields. Verification uses the locally registered expected context. A Fiat–Shamir realisation includes their canonical encodings in its challenge transcript

and must establish the corresponding context-bound argument-of-knowledge property.

Current-state assumption. The authenticated service $\text{CurrentState}(ns)$ returns the latest authorised state at a well-defined read point in a consistent ordering of revocation updates. The verifier reads this service when creating a request and again immediately before its final acceptance check. The latter read defines the logical verification epoch. An update ordered after that read does not retrospectively invalidate the decision. Verification fails if the service is unavailable, its response cannot be authenticated, or the request references a different state. A changed state requires an updated witness and a new request. A signature on an old state alone does not establish freshness.

B. Generic PQ-DID Framework

The eight core lifecycle interfaces are

$$\Pi_{\text{PQ-DID}} = (\text{Setup}, \text{DID.KeyGen}, \text{DID.Publish}, \\ \text{DID.Resolve}, \text{Issue}, \text{Present}, \\ \text{Verify}, \text{Revoke}).$$

Issuance is interactive and verification is stateful. Request creation, current-state retrieval and witness updating are supporting procedures.

- **System setup.**

$$(\text{params}, st_I, st_R, rs_0) \\ \leftarrow \text{Setup}(1^\lambda).$$

Setup generates the public parameters, separate issuer and revocation authority states, and the initial authenticated state for an empty revoked set. Secret outputs are delivered only to their owners. Application trust configuration determines which instances and DID methods are accepted. Application configuration, including the schema, issuer reference, registry and audience trust anchors, is fixed before setup and treated as an implicit input to the setup algorithms. The parameter λ indexes the prescribed parameter family; a concrete profile must specify its parameters and their security justification. Setup may return $\perp$ on failure. Unsupported inputs and operations exceeding the configured capacity are rejected; correctness is evaluated on admissible executions.

- **Identifier and holder-state generation.**

$$(did, st_H) \leftarrow \text{DID.KeyGen}(\text{params}).$$

The selected DID-method adapter generates the identifier and controller-authorisation state, together with the credential-binding secret. Omitting a persistent holder authentication key from a DID document does not remove method-specific controller authorisation.

- **DID publication.**

$$(st'_{ctl}, v_D) \\ \leftarrow \text{DID.Publish}(\text{params}, did, \text{meta}, st_{ctl}).$$

The method adapter authorises publication or update and returns the authenticated document/state reference v_D

and updated controller state. The holder updates st_H accordingly. Invalid or unauthorised requests return $\perp$.

- **DID resolution.**

$$(doc, v_D) \leftarrow \text{DID.Resolve}(\text{params}, did, \nu).$$

The selector ν requests the current document or a supported historical state. The adapter validates the result under the selected DID method. Unavailable, unauthenticated or unsupported results produce $\perp$.

- **Interactive credential issuance.** Let $q_{iss} = (\text{params}, \mu, rs_e)$. With party-specific inputs, write

$$(st'_I, st'_R; out_H) \leftarrow \text{Issue} \left[\begin{array}{l} I : (st_I, ev); \\ R : st_R; \\ H : (st_H, did, v_D, attr) \end{array} \right] (q_{iss}).$$

Here ev is the evidence used by the issuer to validate attributes. The authority states remain with their owners, and $out_H = (vc, w_e)$ or $\perp$ is the holder's output. Distinct authorities communicate through their authorised interfaces; the issuer does not receive the manager's secret state.

In an authenticated and confidential issuance session, the issuer validates the attributes, resolves the current DID state, checks its agreement with v_D , and verifies controller authorisation for that state. These checks determine the complete approved message vector m^{app} , including (did, v_D) . The parties then execute Enroll_{DAA} , which binds this vector, the holder secret, the instance metadata and a fresh registered revocation identifier rid . The holder secret and any non-empty private binding-opening material are not supplied to the issuer. The manager provides an initial non-membership witness, and the holder validates the resulting credential and witness before accepting.

Identifiers are never reused, including identifiers allocated in aborted sessions. Every certification response is recorded before release. An abort stops further processing but does not undo a released certificate, an identifier allocation or a committed public-state update. Consequently, updated authority states are returned even when the holder's output is $\perp$.

The issuer observes the submitted evidence, approved vector, binding representation and its protocol transcript. The manager observes its registration inputs. These private records are not disclosed to verifiers under the baseline trust model.

- **Presentation.**

$$vp \leftarrow \text{Present}(\text{params}, vc, st_H, w_e, ctx, rs_e).$$

The holder validates the authenticated request, policy approval, instance and state references, credential and witness. It invokes Auth_{DAA} to prove the complete relation defined in Section V-B, returning $\perp$ on failure. The presentation binds the disclosed values to the same

certified messages and holder secret, and proves non-revocation of the same certified rid . The DID is disclosed only if required by the approved policy.

- **Stateful verification.**

$$(b, st'_V) \leftarrow \text{Verify}(\text{params}, vp, ctx, rs_e, st_V).$$

The verifier checks that the expected context is registered, pending, unexpired, intended for its audience and associated with the invoking session. It checks issuer trust, the pinned key version, schema and policy, state authentication and Verify_{DAA} . Any required credential-validity fields must be certified and disclosed. If the policy requires a disclosed DID-document check, the verifier resolves the specified state and enforces that check.

After these checks, it reads $\text{CurrentState}(\text{ns})$ and requires agreement with ref_e . It then atomically rechecks the pending context and its expiry and consumes the challenge before returning $b = 1$. All other outcomes return $b = 0$. Trusted time supports expiry checks. A challenge is accepted at most once, including across concurrent requests. The revocation epoch is the final read point defined in Section IV-A.

- **Revocation.**

$$(st'_R, rs_{e+1}, u_{e+1}) \leftarrow \text{Revoke}(\text{params}, st_R, rs_e, rid, req).$$

The manager checks the request's authorisation, the identifier's registration and namespace, and consistency with its current state. For an authorised request concerning an unrevoked identifier, it sets $\mathcal{R}_{e+1} = \mathcal{R}_e \cup \{rid\}$ and publishes the new authenticated state and public update information u_{e+1} . The update authenticates the namespace, consecutive epochs and old and new state values. Invalid, unauthorised or duplicate requests return $\perp$ without changing the state. All information actually published by this procedure is included in the privacy experiments.

C. Witness Updates and Correctness

The holder updates its witness locally using the auxiliary interface

$$w_{e'} \leftarrow \text{UpdateWit}(pp, rid, w_e, rs_e, rs_{e'}, U_{e \rightarrow e'}),$$

where $e' \geq e$ and $U_{e \rightarrow e'} = (u_{e+1}, \dots, u_{e'})$ contains the ordered public updates. Neither the holder's private identifier and witness nor a manager secret key is published or supplied to a remote service by this local procedure. It checks the authenticated update chain, its endpoints and the input and resulting witnesses. Missing or inconsistent updates, or invalid witnesses, yield $\perp$. For $e' = e$, the empty update preserves a valid witness.

Let $\text{NRVerify}(pp, rs_e, rid, w_e)$ be the domain-checked non-membership predicate specified in Section V-B. For a valid

initial witness, honestly generated updates and $rid \notin \mathcal{R}_{e'}$, correctness requires

$$\Pr[\text{NRVerify}(pp, rs_{e'}, rid, w_{e'}) = 1] \geq 1 - \text{negl}(\lambda).$$

For a revoked identifier, honest updating returns $\perp$. Resistance to a maliciously constructed non-membership witness is a separate security property.

Lifecycle correctness. Consider an honest execution with successful issuance, an accepted issuer instance, a satisfied supported policy, an eligible unrevoked credential and a valid updated witness. The state must match the context and the final current-state read. The context must remain pending and unexpired, with no competing acceptance consuming it, and any required DID and application checks must succeed. For vp generated by Present on these inputs and (b, st'_V) generated by Verify, correctness requires

$$\Pr[b = 1] \geq 1 - \text{negl}(\lambda).$$

The probability is over honest setup and protocol randomness. Acceptance of stale states, revoked credentials or unsatisfied policies is not a correctness requirement.

D. DID Binding and Security Objectives

DID binding gives a credential issuance-time controller-authorisation semantics. The issuer validates control of (did, v_D) in the same session in which that reference and the approved attributes are bound to the holder secret. Later presentations prove consistency with that certified statement while concealing the DID. This links issuance to the selected method's identifier-management rules [13]; it does not establish current control of a hidden DID after a controller change. Such changes require an application policy for revocation or reissuance. Current hidden-DID control would require an additional authenticated current-state relation.

Section VI formalises authorised certification, preservation of the certified holder binding, knowledge-based authentication soundness, adaptive presentation privacy, revocation soundness and post-revocation unlinkability. Session binding and one-time challenge consumption supply replay resistance at the lifecycle layer. Privacy is relative to authorised public disclosures and the permitted corruption model. Post-revocation unlinkability protects historical presentations against later public revocation information while holder secrets remain uncompromised; it is distinct from forward security after secret-key exposure.

V. PQ-DAA: A PRIVACY-PRESERVING AUTHENTICATION LAYER FOR PQ-DID

PQ-DAA supplies the authentication interfaces used by $\text{Present}_{\text{DID}}$ and $\text{Verify}_{\text{DID}}$. Its setup, enrolment and presentation syntax follows the anonymous-credential setting, including lattice-based constructions [24], [27]. The four interface names do not themselves define a new cryptographic primitive. Here the layer specifies the joint requirements linking issuer-approved messages, a holder secret, issuance-time DID binding, context-bound selective disclosure and hidden-identifier non-revocation within the PQ-DID lifecycle.

A. Credential State and Disclosure Policies

The holder-local credential record is

$$cred = (\text{cert}, m, rid, \rho, \mu).$$

Here cert is the issuer-certification material, $m = (m_1, \dots, m_\ell)$ is the complete certified vector, rid is its revocation identifier, and ρ is any additional private binding-opening material, possibly ϵ . Setting $\rho = \epsilon$ does not remove the requirement to know x_H . The vector includes the application attributes and designated fields for (did, v_D) . The instance metadata μ and secret x_H are as defined in Section IV-A. The controller state st_{ctl} supports DID operations and issuance validation; it is not an authentication-proof input.

The baseline retains the certified credential record between presentations. Fresh presentation randomness does not modify that record. The epoch-dependent witness w_e is stored separately, giving the authentication witness

$$w_e = (\text{cert}, x_H, m, rid, \rho, w_e).$$

These are private holder records, not public credential or presentation encodings.

A baseline policy is $policy = (D, P_{\text{pub}})$, where $D \subseteq \{1, \dots, \ell\}$ is the canonically ordered set of disclosed fields and P_{pub} checks their values m_D and public context. Policies belong to the family fixed by pp ; Section VII selects disclosures and public equality or range checks on disclosed fields. DID-field disclosure requires holder approval. Any validity fields used by a public check must be certified and disclosed. Disclosing an issuer-certified eligibility flag is supported; proving a range condition on a hidden value requires an explicit additional relation, encoding and proof. Arbitrary hidden-attribute predicates are not implied by selective disclosure.

B. Complete Authentication Relation

The following deterministic, domain-checking interfaces specify the credential and revocation relations. Each instantiation must provide their algorithms and valid encodings.

- $\text{BindRep}(pp, \mu, \text{cert}, \rho)$ returns the original certified binding representation B , or $\perp$. The issuer observes B during enrolment, but B need not appear in a presentation. This interface does not require B to hide the approved attributes from the issuer.
- $\text{BindOpen}(pp, B, x_H, m, \rho)$ checks that B binds the stated secret and complete message vector using ρ , including their domains, encodings and any required witness bounds.
- $\text{CredValid}(pp, \mu, \text{cert}, x_H, m, rid, \rho)$ checks issuer certification of the protocol suite, metadata, binding representation B and identifier rid , together with $\text{BindOpen}(pp, B, x_H, m, \rho) = 1$ for that same B . It checks all credential domains but does not access the issuer's private authorisation records. Whether the messages were actually approved is addressed by certification security.
- $\text{NRVerify}(pp, rs_e, rid, w_e)$ checks the revocation identifier and witness domains and non-membership against A_e . The interface accommodates an accumulator or an

authenticated revocation tree; it does not require constant-size witnesses. State authentication is checked separately below; state freshness is enforced by the PQ-DID wrapper.

The base public statement is

$$X = (pp, \mu, ctx, rs_e, D, m_D).$$

Define $\text{PubOK}(X) = 1$ exactly when parsing and public-domain checks succeed; the suite, protocol tag, issuer/key/schema reference and namespace match the configured instance; (D, P_{pub}) is the supported policy in ctx ; disclosures have the prescribed ordering and encodings; the context's state reference matches rs_e ; and $\text{StateAuth}(pp, rs_e) = 1$ under the configured revocation key. Application trust decisions, trusted time, current-state reads, pending-session checks and holder approval remain wrapper operations.

The complete relation $\mathcal{R}_{\text{auth}}(X, \omega_e) = 1$ holds if and only if

$$\begin{aligned} \text{PubOK}(X) &= 1, \\ \text{CredValid}(pp, \mu, \text{cert}, x_H, m, rid, \rho) &= 1, \\ \text{NRVerify}(pp, rs_e, rid, w_e) &= 1, \\ \text{proj}_D(m) &= m_D, \\ P_{\text{pub}}(m_D, ctx) &= 1. \end{aligned}$$

The identical rid and certified message vector occur throughout. Separate proofs about unrelated certification and revocation witnesses do not satisfy this relation.

An instantiation may introduce additional public proof data a and private randomness η , with a refined relation satisfying

$$\mathcal{R}_{\text{inst}}((X, a), (\omega_e, \eta)) \implies \mathcal{R}_{\text{auth}}(X, \omega_e).$$

It must give an efficient honest procedure lifting a valid base witness to this relation, together with the applicable completeness, privacy and extraction guarantees. Auxiliary public data are covered by the privacy experiment and bound into the statement with ctx . Section VII requires no auxiliary relation data: $a = \eta = \epsilon$. Its proof nonce, view commitments and opened views are part of π , and its random shares and tapes are prover randomness rather than additional credential witnesses.

C. Generic PQ-DAA Syntax

The four core interfaces are

$$\Pi_{\text{PQ-DAA}} = (\text{Setup}, \text{Enroll}, \text{Auth}, \text{Verify}).$$

Subscripts distinguish the DID and DAA layers when necessary.

- **Setup.**

$$(pp, st_I, st_R, rs_0) \leftarrow \text{Setup}(1^\lambda).$$

This generates the cryptographic parameters, separate authority states and initial authenticated revocation state, or returns $\perp$ on setup failure. PQ-DID incorporates these same outputs into its setup; it does not generate a second, independent credential instance.

- **Interactive enrolment.** For $q_{\text{en}} = (pp, \mu, rs_e)$ and the issuer's approved vector m^{app} , write

$$(st'_I, st'_R; out_H) \leftarrow \text{Enroll} \left[\begin{array}{l} I : (st_I, m^{\text{app}}); \\ R : st_R; \\ H : (x_H, m^{\text{app}}) \end{array} \right] (q_{\text{en}}).$$

The holder constructs B and any private opening material ρ , and the manager allocates a fresh registered rid . The holder proves knowledge of (x_H, ρ) satisfying

$$\text{BindOpen}(pp, B, x_H, m^{\text{app}}, \rho) = 1$$

in an enrolment statement binding $(pp, \mu, m^{\text{app}}, rid, B, n_I, rs_e)$. Here n_I is a fresh, one-use issuer challenge. The issuer verifies its own approved vector, the instance and state references, the identifier's registration, the active challenge and the proof before releasing certification of the suite, μ , B and rid .

The issuer records the certification before release and consumes the issuance challenge. The protocol does not transmit x_H or non-empty private opening material to the issuer; the selected enrolment argument must provide its specified witness-privacy guarantee. The holder checks the certificate and initial witness and returns $out_H = (cred, w_e)$ on success or $\perp$ otherwise. The allocation, logging and abort semantics are those of Section IV-B. PQ-DID supplies m^{app} after external attribute and controller validation; enrolment proves binding to that vector. Certification security follows either a direct reduction or the compatible issuance-extraction route of Section VI-B3. The construction in Section VII uses direct certification of the issuer-approved encoded vector.

- **Authentication.**

$$auth \leftarrow \text{Auth}(pp, cred, x_H, w_e, ctx, rs_e).$$

After holder approval and input validation, the algorithm constructs X , checks the complete relation and generates a fresh proof π , or returns $\perp$. Its output is

$$auth = (D, m_D, \pi).$$

If auxiliary public data are required, $\pi = (a, \pi_{\text{inst}})$. Authentication leaves the certified record unchanged and does not write to issuer or revocation-manager records. Reusability means that the same credential can support new contexts using fresh proof randomness and appropriately updated witnesses.

- **Cryptographic verification.**

$$b \leftarrow \text{Verify}(pp, auth, ctx, rs_e).$$

The verifier derives μ from the expected context and state, constructs X , checks $\text{PubOK}(X)$ and $P_{\text{pub}}(m_D, ctx)$, and verifies the complete-relation proof. Malformed encodings, unsupported policies, mismatched disclosures and invalid proofs cause rejection.

This algorithm is stateless: it checks cryptographic validity for the supplied statement. PQ-DID supplies issuer

trust, current-state freshness, trusted-time and session checks, and one-time challenge consumption.

D. Revocation and Composition with PQ-DID

Revocation uses Section IV-B, and local witness updating uses Section IV-C. An authentication for $rs_{e'}$ uses a valid $w_{e'}$ and a context referencing that state. Updating the witness does not change the certified messages, binding representation or identifier. Honest authentication fails when no valid witness is available.

Issue_{DID} validates the external evidence and DID control before invoking Enroll_{DAA}. Its local record vc contains $cred$ and any additional lifecycle metadata. Present_{DID} validates the request and holder approval and invokes Auth_{DAA} with the appropriate witness. Verify_{DID} invokes Verify_{DAA} within its stateful procedure. Challenge consumption occurs only in the DID layer.

E. Correctness and Security Requirements

For an honest accepted enrolment, a valid witness in the authenticated state, a satisfied supported policy and matching context references, let $auth$ be the output of Auth. Correctness requires

$$\Pr[\text{Verify}(pp, auth, ctx, rs_e) = 1] \geq 1 - \text{negl}(\lambda).$$

Lifecycle acceptance additionally requires the conditions in Section IV-C.

Section VI separates the following requirements. *Authorised certification* prevents a valid credential opening outside the issuer's recorded approvals. *Binding preservation* prevents different holder-secret/message pairs from opening the same certified representation. *Knowledge soundness* requires an applicable quantum extractor for fresh-context proofs of the complete relation; it does not posit extraction from a single transcript. *Adaptive presentation privacy* covers anonymity, unlinkability and selective-disclosure privacy under repeated presentations and permitted verifier collusion. *Revocation soundness* excludes valid non-membership witnesses for revoked identifiers. *Post-revocation unlinkability* additionally covers the joint view of historical tokens and subsequent public revocation information. These guarantees do not prevent voluntary sharing of a complete credential and its holder secret.

VI. GENERIC SECURITY MODEL OF THE PQ-DAA AND PQ-DID FRAMEWORKS

We define security for the interfaces of Sections IV and V. The model separates authorised certification, preservation of the holder binding, complete-witness knowledge, adaptive presentation privacy and revocation security. Here Π denotes a PQ-DAA instance with its enrolment, certification and revocation interfaces. PQ-DID experiments additionally execute the lifecycle wrapper.

The asymptotic definitions concern the parameter family indexed by λ . A fixed profile, such as Section VII, must give concrete bounds for its specified parameters and adversarial resources. Choosing $\lambda = 128$ for a nonce convention does not establish an overall 128-bit security claim.

A. Execution Model, Oracles and Leakage

Adversaries are quantum polynomial-time (QPT). Protocol messages and protocol-oracle queries are classical. For a Fiat–Shamir realisation, the adversary additionally has quantum superposition access to the random oracle. Probabilities include setup, protocol, adversarial and measurement randomness. All experiments allow polynomially many adaptive queries. Message, metadata and binding comparisons use the canonical encodings fixed by pp . Unsupported or capacity-exceeding operations follow the specified rejection or failure behaviour.

The baseline uses sequential protocol-oracle scheduling: an interactive enrolment completes or aborts before another protocol-oracle call begins. Quantum random-oracle queries remain available during that interaction. Thus the model does not assert security for interleaved malicious enrolments. Several verifier contexts may nevertheless remain pending, and the lifecycle interface retains atomic challenge consumption.

The challenger maintains holder handles h , credential or registration handles i , the private issuer-authorisation table $\mathcal{T}_{\text{auth}}$, authenticated revocation history $\{rs_e, u_e\}_e$, and an ordered execution history $\mathcal{J}$. Handles name experiment records and are not embedded in anonymous tokens. Records distinguish allocated identifiers, released certification responses and successfully accepted holder credentials. Every allocation and certification response remains recorded after an abort.

Corruption model. Setup, the targeted issuer instance, its revocation manager and the current-state service remain honest. Their private records are not revealed to the adversary. DID-method integrity and authenticated resolution are assumed. Verifiers may collude with each other and with corrupted holders; a designated target verifier remains honest in the PQ-DID acceptance experiment. Every holder may be corrupted in certification, binding and non-membership experiments.

Privacy protects only holders that remain uncorrupted throughout the experiment, including after revocation. Corrupting a holder exposes all its credentials and retained state. Authority–verifier collusion, authority compromise and challenge-holder secret compromise are not covered by these privacy claims.

The experiment provides the following oracles. Malformed queries, unknown handles and inadmissible operations return $\perp$ without using the hidden challenge bit.

- $\mathcal{O}_{\text{Enroll}}(h, m, \mu)$ executes the prescribed issuance procedure. A new holder is designated honest or adversarially controlled before the session. The issuer applies the configured evidence and authorisation checks; the requested vector m is not automatically approved. For an honest holder, the challenger retains its private inputs and executes the protocol. For a controlled holder, the adversary supplies its inputs and interacts with the honest issuer. The oracle returns the handles, outcome, public outputs and the adversary's own view where applicable. It does not otherwise release honest-holder issuance transcripts or private evidence. Every released certification response is entered in $\mathcal{T}_{\text{auth}}$ before release, whether or not the holder subsequently accepts it.

- $\mathcal{O}_{\text{Show}}(i, ctx)$ requests a presentation from a named, honestly issued credential. The challenger checks holder approval, the supported policy, current state, credential eligibility and a fresh presentation context. It updates the private witness when necessary and runs Auth with fresh randomness. It returns the token and public inputs, or $\perp$ on failure. Repeated use of a credential under new contexts is allowed. Every returned base statement X is recorded in $\mathcal{Q}_{\text{show}}$, together with the token. A context is fresh here if its audience/session/nonce combination has not already been used by a presentation oracle.
- $\mathcal{O}_{\text{Revoke}}(i)$ requests revocation of a named registered identifier. The experiment authorises the request unless a challenge restriction forbids it, including for an identifier allocated in an aborted session. It returns the actual published state and update records, including any identifier or metadata published by the selected construction. This experimental capability does not confer revocation authority on an arbitrary application user.
- $\mathcal{O}_{\text{Update}}(i, e')$ executes local witness updating through a valid ordered chain from the retained witness epoch to an available later epoch. For an honest holder, it retains the witness privately and returns only the public outcome. For a controlled or corrupted holder, the resulting witness is available to the adversary. All public update records remain available independently of this oracle.
- $\mathcal{O}_{\text{Corrupt}}(h)$ releases the holder's retained secrets, credentials, witnesses, controller state and local records. A protected challenge holder cannot be corrupted; a previously corrupted holder cannot become a challenge holder.
- In PQ-DID experiments, $\mathcal{O}_{\text{Request}}$ creates an honest verifier's pending context using the configured trust policy and current-state service. $\mathcal{O}_{\text{Verify}}$ executes stateful verification and updates verifier state. Authenticated public resolution queries and their outputs are included in the execution history. PQ-DAA verification is public and stateless.

Public leakage. The authorised presentation leakage is

$$L = (\mu, ctx, rs_e, D, m_D),$$

together with public parameters and the public directory and issuance metadata explicitly fixed by the selected deployment profile. The adversarial view contains every permitted oracle reply, all observed proof bytes and lengths, and the adversary's residual quantum state. Additional proof data are not automatically classified as authorised leakage; their joint distribution must satisfy the privacy experiment. A deployment cannot silently add credential-dependent public fields to this leakage specification. In Section VII, the complete proof bytes remain observable, while the certified binding and revocation witness remain private. Updates publish the revoked identifier and its old sibling path; these publications belong to the adaptive and historical privacy views.

Supported policies may change adaptively between sessions, subject to holder approval and the challenge restrictions. Privacy is relative to the cumulative authorised disclosures, which can themselves identify a holder. A hidden-predicate

extension must specify the relation and add its public outcome to the leakage. Network addresses, traffic timing and service availability are outside these credential-level experiments.

B. Certification, Binding and Knowledge Requirements

1) *Unforgeability of Authorised Credentials:* For every released certification response, the issuer records

$$t = (\mu, m^{\text{app}}, rid, B)$$

in $\mathcal{T}_{\text{auth}}$. The complete approved vector includes the certified DID reference. The issuer need not know the holder secret to maintain this table. Let

$$z = (\mu, \text{cert}, x_H, m, rid, \rho)$$

be a credential opening. For $t = (\mu_t, m_t, rid_t, B_t)$, define $\text{Match}(t, z) = 1$ if and only if

$$\mu = \mu_t, \quad m = m_t, \quad rid = rid_t,$$

$$\text{BindRep}(pp, \mu, \text{cert}, \rho) = B_t.$$

Thus matching is defined through the original certified binding, not a freshly generated presentation commitment. Set

$$\text{Authorised}(z, \mathcal{T}_{\text{auth}}) = \bigvee_{t \in \mathcal{T}_{\text{auth}}} \text{Match}(t, z).$$

In $\text{Exp}_{\Pi}^{\text{cert}}(\mathcal{A})$, the adversary has the applicable oracles with arbitrary holder corruptions and outputs z^* . It wins when μ^* identifies the targeted instance and

$$\text{CredValid}(pp, \mu^*, \text{cert}^*, x_H^*, m^*, rid^*, \rho^*) = 1,$$

$$\text{Authorised}(z^*, \mathcal{T}_{\text{auth}}) = 0.$$

Define $\text{Adv}_{\Pi}^{\text{cert}}(\mathcal{A})$ as the probability of winning. Authorised certification requires this advantage to be negligible. Fresh presentation randomness for an authorised record is not a forgery.

This definition links certification to the values actually approved by the issuer. Signing an unchecked holder-supplied binding does not establish that its contents match those approvals. The reduction must use either a direct encoding linkage or an adequate issuance proof. The truth of external evidence remains an issuer-validation assumption.

2) *Preservation of the Holder Binding:* The authorisation table records approved messages and a representation B , but does not record the holder's secret. We therefore state binding preservation separately. In $\text{Exp}_{\Pi}^{\text{bind}}(\mathcal{A})$, honest setup is followed by the applicable oracles with arbitrary holder corruptions. The adversary outputs B and two opening tuples (x_0, m_0, ρ_0) and (x_1, m_1, ρ_1) . It wins if

$$\text{BindOpen}(pp, B, x_0, m_0, \rho_0) = 1,$$

$$\text{BindOpen}(pp, B, x_1, m_1, \rho_1) = 1,$$

$$(x_0, m_0) \neq (x_1, m_1).$$

The winning probability $\text{Adv}_{\Pi}^{\text{bind}}(\mathcal{A})$ must be negligible. The property covers every accepted message domain and opening, including maliciously selected values; it does not assume a particular commitment construction. It remains meaningful when $\rho_0 = \rho_1 = \epsilon$.

Together with validated enrolment, this requirement prevents an efficiently produced alternative secret/message opening for the same certified representation. It does not prevent a holder from voluntarily sharing its original secret and complete credential.

3) *Certification Reduction and Optional Issuance Extraction*: The certification analysis must account for accepted malicious enrolments. A direct reduction is sufficient when the certified representation and its canonical encoding enforce the exact approved vector. Section VII follows this route: the issuer builds the message component of B from m^{app} , records it, and signs it with the instance and identifier. Certification security then concerns a previously unsigned credential message; it does not require extracting every enrolment witness. This does not remove the enrolment proof from the protocol.

For a different reduction that compares a later opening with an opening extracted during issuance, the following is a sufficient extraction contract.

There is an efficient quantum emulation of the permitted sequential issuance execution which retains an analysis-only table $\mathcal{T}_{\text{ext}}$. For every released record $t = (\mu, m^{\text{app}}, \text{rid}, B)$, this table contains (x_H, ρ) satisfying

$$\text{BindOpen}(pp, B, x_H, m^{\text{app}}, \rho) = 1.$$

The joint distribution of the observable execution, authorisation history and adversary's residual quantum state is computationally indistinguishable from the real execution. Missing or invalid extracted entries occur with negligible probability. These openings are neither delivered to the issuer nor revealed to the adversary in the real protocol.

Let $\epsilon_{\text{iss}}(\lambda, q)$ bound the relevant distinguishing loss, extraction failures and any running-time truncation for the polynomial query budget q . It must be negligible. The emulation must preserve the external signing, registration and revocation interactions used by the component reduction; it cannot undo a released certificate or assume access to the inverse of an external classical oracle.

A compatible quantum witness-extended emulation can establish this contract. Sequential scheduling and ordinary knowledge soundness alone do not imply it. An alternative certification proof may avoid this contract if it directly establishes the certification experiment under its stated component assumptions. In either case the precise reduction, permitted oracle access and loss must be supplied.

4) *Complete-Witness Knowledge and Context Binding*: Knowledge soundness concerns the complete relation $\mathcal{R}_{\text{auth}}$ of Section V-B. A refined proof relation must extract a witness that projects to

$$w_e = (\text{cert}, x_H, m, \text{rid}, \rho, w_e)$$

satisfying that relation, including all domains and the identical certified identifier in non-membership verification.

For extraction, a base statement X is fresh when it has not previously been answered by an honest presentation oracle, including the challenge oracle below. Changing only auxiliary proof data a does not make a previously answered base statement fresh. Previously obtained proofs for other contexts remain available as auxiliary information.

We require an applicable quantum argument-of-knowledge theorem for this oracle environment. To state the quantitative contract, let S be a specified efficient target predicate on the recorded history and public statement. Write ϵ_S for the probability that the adversary outputs an accepting proof for a fresh X satisfying S . An extractor with the stated access to that adversary and its oracles must output a consistent history and (X, ω_e) with probability at least

$$F_{\text{auth}}(\epsilon_S; \lambda, q),$$

such that X is fresh, S holds and $\mathcal{R}_{\text{auth}}(X, \omega_e) = 1$. The target predicate and the corresponding adaptive-statement guarantee must be justified by the applicable theorem; they are not inferred from unrestricted extraction. The case $S \equiv 1$ gives the ordinary fresh-statement requirement.

The selected profile must specify the knowledge error κ , the function F_{auth} , the query dependence and the extractor's running time. The bound must be non-negligible whenever $\epsilon_S - \kappa$ is non-negligible, within the permitted running-time model. Any expected-time extraction used in a polynomial-time component reduction must include truncation and its loss.

Extraction used below must be compatible with the actual authorisation and revocation history and the external component oracles. It cannot create an unrelated issuance history or assume the ability to rewind or invert an external classical signing or revocation oracle. If honest proofs are simulated in a reduction, the extraction theorem must cover that simulated-proof access or the proof must justify another route.

The construction analysis must decode its circuit witness ξ into ω_e and derive F_{auth} from its component extraction bound, preserving the same target event and authorised history.

The expected context and all proof-associated public data are bound to the statement. Fiat-Shamir hashing of a context is not, by itself, an extraction theorem. QROM transformation results require an appropriate underlying quantum proof of knowledge and their other hypotheses [34]; they do not automatically establish the issuance-emulation contract above. No claim is made that a witness can generally be recovered from one accepted transcript alone.

C. Adaptive Presentation Privacy

The challenger samples one bit $b \leftarrow \{0, 1\}$ and provides the preceding oracles together with

$$\mathcal{O}_{\text{LR}}(i_0, i_1, \text{ctx}).$$

An admissible query satisfies all the following conditions:

- 1) Both records are accepted honest credentials whose holders have never been corrupted. Both holders become protected for the remainder of the experiment. The indices may coincide.
- 2) Both credentials are eligible and unrevoked in the same current authenticated state, have valid updated witnesses, and satisfy the same supported policy approved by both holders.
- 3) Their instance metadata, disclosed positions and disclosed values agree. The context and every authorised

public outcome are identical for the two candidates, and the context is fresh for the presentation oracles.

The challenger checks both candidates and performs any required local witness maintenance without depending on b . An inadmissible query returns $\perp$. For an admissible query, it returns the public inputs and

$$auth \leftarrow \text{Auth}(pp, cred_{i_b}, x_{H(i_b)}, w_{i_b, e}, ctx, rs_e)$$

using fresh randomness, and records the base statement in Q_{show} . The selected index is not returned, passed to an authority or used to modify credential or revocation records.

The adversary may request polynomially many adaptive presentations across colluding verifiers, change supported policies and interleave background revocations and witness updates. Named presentations from challenge credentials through Q_{show} remain available under other fresh contexts. Candidate credentials cannot be revoked while this challenge oracle remains available. Background holders may be corrupted before or after challenge queries.

Finally, the adversary outputs b' . Define

$$\text{Adv}_{\Pi}^{\text{priv}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

Adaptive presentation privacy requires negligible advantage for every admissible QPT adversary.

Anonymity restricts the candidates to different protected holders. Unlinkability compares repeated use of one credential with use of different eligible credentials; successive pairs (i_0, i_0) and (i_0, i_1) give the usual two-presentation comparison. Selective-disclosure privacy compares different hidden certified values with identical authorised disclosures. These are restrictions of the adaptive experiment, not consequences of a single-token test. All restrictions apply across the complete query history. Revealing a distinguishing DID or other attribute makes that comparison inadmissible; other identifying disclosures can defeat anonymity even when the DID is hidden.

D. Revocation Security and Historical Privacy

1) *Adaptive Non-Membership Soundness*: In $\text{Exp}_{\Pi}^{\text{nm}}(\mathcal{A})$, the adversary has the applicable enrolment, presentation, corruption, revocation and update oracles with no protected holders. It may retain all old states and witnesses. It outputs (e^*, rid^*, w^*) and wins if rs_{e^*} belongs to the generated authenticated history and

$$rid^* \in \mathcal{R}_{e^*}, \\ \text{NRVerify}(pp, rs_{e^*}, rid^*, w^*) = 1.$$

The winning probability $\text{Adv}_{\Pi}^{\text{nm}}(\mathcal{A})$ must be negligible. A valid witness for an epoch before revocation does not violate this definition.

For authentication, the relevant identifier is the one in the extracted certified witness. An adversary's unauthenticated claim that a token belongs to a named credential does not determine its hidden rid . Successful authentication with a different, genuinely unrevoked credential is not a revocation forgery. PQ-DID fixes the applicable epoch through its final current-state read.

2) *Post-Revocation Unlinkability*: The experiment $\text{Exp}_{\Pi}^{\text{pru}}(\mathcal{A})$ extends adaptive presentation privacy in two phases.

- 1) **Historical presentations**. The adversary uses the same global challenge bit and oracles, obtaining challenge tokens across supported policies, verifiers and epochs. All candidate credentials remain unrevoked while challenge queries are available.
- 2) **Subsequent revocation**. At an adaptively chosen cut-off, the challenge oracle closes permanently. The adversary may revoke candidate credentials and receives every actual published state, update and revocation metadata item. At least one credential used in a challenge must be revoked before the final guess. Ordinary enrolment, named presentation, update and permitted corruption queries remain available. Challenge-holder corruption remains forbidden.

Revocation identifies a named handle, with the same service rule in both worlds. Its choice may depend on observations but is not selected by the challenger using b . After the cut-off, no oracle presents again using the hidden challenge-selected index; that would disclose revoked-versus-unrevoked status directly.

The final adversarial view is the joint distribution

$$\text{View}_b = (\text{Hist}_b, \text{RevPub}_{>\text{cut}}, \text{Aux}_b),$$

including all historical tokens, subsequent public revocation information, other permitted replies, corrupted background holders' witnesses and the adversary's residual quantum state. Define

$$\text{Adv}_{\Pi}^{\text{pru}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right|.$$

Post-revocation unlinkability requires negligible advantage for all admissible adversaries. It protects the association of historical presentations with later revocation records; it does not conceal revocation facts deliberately published by the mechanism.

Sufficient joint-view simulation condition. One sufficient proof route is an online simulator for the complete challenge token, including auxiliary public proof data. The simulator receives its public statement and authorised leakage, and any simulation setup permitted by the proof system, but neither the selected witness nor the hidden candidate index. It has no advance knowledge of future revocation requests. For each b , require

$$\text{View}_b^{\text{real}} \approx_q \text{View}_b^{\text{sim}},$$

where the entire subsequent service execution retains its prescribed joint distribution with the honestly generated records. The condition includes the shared random oracle, where applicable. A direct joint-view hybrid proof is an alternative way to establish the experiment; this simulation condition is sufficient, not necessary. Absence of a literally repeated identifier in a later update is insufficient to establish either route.

Theorem 1 (Conditional post-revocation unlinkability). *Suppose the joint-view simulation condition holds, challenge-token generation has no selection-dependent persistent-state effects,*

and all remaining services operate on named records and public requests without receiving the hidden selection. If δ_b bounds the distinguishing gap between the complete real and simulated views in world b , then

$$\text{Adv}_{\Pi}^{\text{pru}}(\mathcal{A}) \leq \frac{\delta_0 + \delta_1}{2}.$$

Proof. Replace challenge tokens online by their simulations in each world. The respective changes in any final binary test are at most δ_0 and δ_1 , including after later revocations. Both simulated worlds provide identical inputs to the simulator, initialise all named records by the same rule, and execute the same service algorithms without access to b . Induction over the adversary's adaptive queries gives identical simulated-view distributions. The triangle inequality bounds the difference between the two real-world guess probabilities by $\delta_0 + \delta_1$. Guessing advantage is half that difference. $\square$

E. Conditional Security of PQ-DID

The following results assume the configured instance and trust model, authenticated resolution and state references, correct external issuance validation, the current-state service, trusted time and atomic challenge management. These are explicit system assumptions; signature unforgeability alone does not supply currentness or external validation.

Theorem 2 (Authentication and revocation soundness). *Suppose the complete-relation extraction contract applies to the PQ-DID acceptance event with success probability ϵ on a previously unanswered base statement. Let p_{good} be the probability that this extraction returns a valid witness whose credential opening is issuer-authorized and whose certified identifier is unrevoked at the recorded final verification epoch. There are compatible component adversaries $\mathcal{B}_{\text{cert}}$ and $\mathcal{B}_{\text{nm}}$ such that*

$$\begin{aligned} p_{\text{good}} &\geq F_{\text{auth}}(\epsilon; \lambda, q) \\ &\quad - \text{Adv}_{\Pi}^{\text{cert}}(\mathcal{B}_{\text{cert}}) \\ &\quad - \text{Adv}_{\Pi}^{\text{nm}}(\mathcal{B}_{\text{nm}}). \end{aligned}$$

All extraction, emulation and running-time losses must be included in the invoked contract or component bounds.

Proof. PQ-DID acceptance implies verification of the complete relation under the expected context and the authoritative state at the final read. Apply the compatible extraction theorem for this target event. It produces a relation witness with probability at least $F_{\text{auth}}(\epsilon; \lambda, q)$.

If the extracted opening does not match $\mathcal{T}_{\text{auth}}$, output it in the certification experiment. If the extracted *rid* belongs to the revoked set at the recorded epoch, output that epoch, identifier and witness in the non-membership experiment. Both reductions preserve the authorisation and revocation history required by their games. By the union bound, subtracting these two bad-event probabilities leaves the stated lower bound on authorised, unrevoked extraction. $\square$

The bound concerns successful knowledge extraction, not an algorithm which extracts from each individual accepted transcript. Binding preservation separately excludes two efficiently produced distinct secret/message openings for the same

certified representation. A reduction that relies on extracting a malicious enrolment opening must include the corresponding issuance-emulation and extraction losses in Section VI-B3. The direct-certification route does not incur those particular losses; authentication extraction and all other component losses remain. The external validation assumption gives the resulting certified DID reference its issuance-time controller-authorisation meaning.

The stateful verifier rejects expired, consumed or inappropriate contexts and stale revocation references. Atomic challenge consumption and the non-reuse of nonces prevent a second acceptance of the same request. These checks do not establish resistance to voluntary secret sharing or live relay of an authorised response within its intended session.

Theorem 3 (Privacy under lifecycle composition). *Let $g \in \{\text{priv}, \text{pru}\}$ denote the corresponding PQ-DAA privacy experiment. Suppose Π satisfies that experiment, and all additional PQ-DID fields and observable service effects admit online simulation from its authorised leakage and permitted public history. Let $\Delta_{\text{wrap},b}$ bound the change in the complete world- b view due to this wrapper simulation. Then an admissible PQ-DID adversary $\mathcal{A}$ yields a PQ-DAA adversary $\mathcal{B}$ with*

$$\begin{aligned} \text{Adv}_{\text{PQ-DID}}^g(\mathcal{A}) &\leq \text{Adv}_{\Pi}^g(\mathcal{B}) \\ &\quad + \frac{\Delta_{\text{wrap},0} + \Delta_{\text{wrap},1}}{2}. \end{aligned}$$

For $g = \text{pru}$, the simulation requirement covers the complete future revocation and auxiliary-information view.

Proof. The reduction forwards admissible candidate pairs and contexts to the underlying PQ-DAA experiment and simulates the additional lifecycle observations. These include public DID and issuer-key resolution, metadata, request handling and observable state-service effects. The simulation receives no hidden candidate index. Replacing the wrapper in each world changes the corresponding final test probability by at most $\Delta_{\text{wrap},b}$. The remaining distinguishing advantage is that of $\mathcal{B}$ in the underlying experiment. The triangle inequality and the definition of guessing advantage give the bound. For historical privacy, the same reduction continues through all permitted subsequent revocations. $\square$

Thus the lifecycle result retains the separate dependencies on authorised certification, binding preservation, complete-relation quantum extraction, non-membership soundness, presentation privacy and the system services. These definitions fix the guarantees to be established for the selected construction; they do not replace the component algorithms or their security reductions.

VII. CONCRETE INSTANTIATION OF THE PQ-DID LIFECYCLE

We realise Sections IV-B and V-C using ML-DSA-65, SHA3-384 binding, a revocation tree and a Boolean MPC-in-the-head argument. The reference DID method retains the registry, confidential issuance, current-state, trusted-time and atomic verifier-state assumptions of Section IV-A; it does not specify consensus or a standardised DID method.

Write $H = \text{SHA3-384}$, $[a]_k = \text{I2OSP}(a, k)$ (unsigned big-endian), $\text{LP}(s) = [|s|]_4 \parallel s$, and

$$\text{enc}_a(s_1, \dots, s_k) = \text{LP}(a) \parallel [k]_4 \parallel \bigparallel_{i=1}^k \text{LP}(s_i).$$

E recursively encodes tuples using their literal ASCII tags and displayed field order; lists contain one field per element. Byte strings are inserted once. Reject incorrect tags, counts, types, domains, lengths, padding or trailing bytes; every length-prefixed field must contain at most $L = 2^{32} - 1$ bytes. ML-DSA retains its standard internal byte order.

A. Eight Concrete Lifecycle Algorithms

- 1) **Setup:** $\text{Setup}(1^\lambda)$. Accept $\lambda = 128$ only, with suite PQ-DID-MITH-1. Use August 2024 FIPS 204 ML-DSA-65 with fresh hedging randomness [30] and FIPS 202 SHA3/SHAKE [35]. Public keys, secret keys and signatures occupy 1952, 4032 and 3309 bytes. $\text{Sign}_a, \text{Vfy}_a$ use external context PQ-DID/ $a/v1$, for *credential*, *state*, *update*, *did-record*, *did-read*, *control*, *request*, *current* and *revreq*; subscript *cred* abbreviates *credential*. Generate independent issuer/revocation key pairs and a uniform 32-byte namespace. Set

$$\begin{aligned} pp &= (\text{suite}, \text{ref}_I, \text{ns}, pk_I, pk_R, \text{sch}), \\ \mu &= (\text{ref}_I, \text{ns}), \quad \text{cfg}_D = (\gamma, pk_G), \\ \text{params} &= (pp, \text{cfg}_D, \text{trust}). \end{aligned}$$

The registry has a 32-byte identifier γ , independent ML-DSA keys, a versioned map, atomic append-if-current and ordered reads; it retains sk_G . Trust configuration authorises the issuer/key/schema instance and pins audience request keys. Define $\text{ref}_I = \text{enc}_{\text{iref}}(\text{did}_I, \text{kid}, E(\text{sch}))$, with non-empty issuer/key identifiers of at most 256 bytes. Tags for $pp, \mu, rs_e, \text{ref}_e$ are *parameters*, *meta*, *rstate*, *rref*; repeated instance fields must agree.

The immutable schema comprises $2 \leq \ell \leq 16$ descriptors $(\text{name}_j, \text{type}_j, M_j)$: distinct non-empty ASCII names of at most 64 bytes, one-byte types and two-byte capacities. Types 0, 1 and 2 denote byte strings, one Boolean byte in $\{0, 1\}$ and eight-byte unsigned integers. Tag schema contains the one-byte count, two distinct one-byte indices for (did, v_D) , then descriptors tagged *field*. These designated fields have type 0 and capacities at least 171 and 56 bytes. Require sufficient fixed-type capacity and $\sum_j (2 + M_j) \leq 1024$. For canonical b_j with $|b_j| \leq M_j$, set

$$E_j(m_j) = [|b_j|]_2 \parallel b_j \parallel 0^{8(M_j - |b_j|)}.$$

Concatenating fields and zero-padding gives the 1024-byte $E_{\text{sch}}(m)$. A two-byte mask represents D by bit $j-1$; unused bits are zero, and m_D concatenates selected E_j in ascending order. Tag *policy* contains this mask followed by at most 32 distinct, lexicographically sorted clauses $\text{enc}_{\text{eq}}([j]_1, E_j(c))$ or $\text{enc}_{\text{range}}([j]_1, [a]_8, [b]_8)$. All tested fields are disclosed; ranges require type 2

and $a \leq b$. P_{pub} evaluates their conjunction. Encode Section IV-A's *ctx* under *context*, with *tag* = *suite*, non-empty *aud*, *sid* of at most 256 bytes, a 32-byte nonce and eight-byte UTC expiry in seconds.

Use a depth-20 tree:

$$\begin{aligned} L_b &= H(\text{enc}_{\text{leaf}}(\text{suite}, \mu, b)), \\ F_j(u, v) &= H(\text{enc}_{\text{node}}(\text{suite}, \mu, j, u, v)), \\ T_e[0, i] &= L_{\mathbf{1}[i \in \mathcal{R}_e]}, \\ T_e[j, k] &= F_j(T_e[j-1, 2k], T_e[j-1, 2k+1]), \\ A_e &= T_e[20, 0]. \end{aligned}$$

b, j occupy one byte; hashes occupy 48 bytes. Initially, $\mathcal{R}_0 = \emptyset$, $Z_0 = L_0$, $Z_j = F_j(Z_{j-1}, Z_{j-1})$ and $A_0 = Z_{20}$. For eight-byte epochs without wraparound, set

$$\begin{aligned} \tau_e &\leftarrow \text{Sign}_{\text{state}}(sk_R, \text{enc}_{\text{state}}(\text{suite}, \mu, e, A_e)), \\ rs_e &= (\text{ns}, e, A_e, \tau_e). \end{aligned}$$

StateAuth checks domains, instance and signature. Each authority retains its secret key and initially empty tables: issuer approvals/sessions and manager registrations/requests. The manager retains the tree and allocation counter zero. Validate keys and bounded expansions, sign rs_0 and initialise its current-state register. Return $(\text{params}, st_I, st_R, rs_0)$, or $\perp$ without activating a partial instance. Setup_{DAA} supplies the cryptographic outputs once; PQ-DID adds method/trust configuration.

- 2) **Holder generation:** DID.KeyGen(params). Generate and validate controller keys (pk_C, sk_C) ; independently sample $x_H, \zeta \leftarrow \{0, 1\}^{256}$. Set

$$\begin{aligned} d &= H(\text{enc}_{\text{did-id}}(\text{suite}, \gamma, pk_C, \zeta)), \\ \text{did} &= \text{did:pqdid} : \parallel \text{hex}(\gamma) \parallel : \parallel \text{hex}(d), \end{aligned}$$

using lower-case hexadecimal without prefixes (171 ASCII bytes). Return (did, st_H) , where $st_H = (x_H, st_{\text{ctl}})$ and controller state stores the DID, keys, salt, reference ϵ and no pending operation; failure returns $\perp$. Nothing is published or registered.

- 3) **Publication:** DID.Publish($\text{params}, \text{did}, \text{meta}, st_{\text{ctl}}$). Define

$$\begin{aligned} q_k &= (\gamma, \text{did}, k, \delta_{k-1}, pk_{C,k}, a_k, \zeta), \\ R_k &= (q_k, \alpha_k), \quad \delta_k = H(E(R_k)), \quad v_D = [k]_8 \parallel \delta_k. \end{aligned}$$

Tags are *did-body* and *did-record*; $a_k \in \{0, 1\}$ occupies one byte, and $0 \leq k < 2^{16}$ eight bytes. Metadata $(a, r) \in \{(1, 0), (1, 1), (0, 0)\}$ comprises one-byte active/rotation flags. Genesis requires $(1, 0)$, authenticated absence, $k = 0$, $\delta_{-1} = 0^{384}$ and recomputation of the DID; sign $E(q_0)$ with $sk_{C,0}$ under *did-record*. Updates require a fresh current read matching the retained reference and controller key, an active predecessor, next index, unchanged DID and salt, and the predecessor digest. Sign with the previous controller key; $r = 1$ installs newly generated, validated keys, otherwise retain the key. Deactivation $(0, 0)$ is terminal.

Validate and retain the pending record/keys. The registry checks the record/transition and atomically appends

against its predecessor or absence, preserving history. A confirming authenticated read returns (st'_{ctl}, v_D) , changing only controller state. Resolve pending attempts before new operations: retry identical records while their predecessor is current, accept confirmed occurrence even after deactivation, or retire an attempt upon a conflicting successor. Timeouts retain possibly active keys. Unsupported metadata and invalid operations return $\perp$; committed records survive delivery failure.

- 4) **Resolution:** DID.Resolve(params, did, ν). Validate the DID/method/registry. Selector ν is byte 0 for current state or byte 1 followed by v_D for history. Send a fresh 32-byte n_G ; the pinned registry signs $(\gamma, did, \nu, n_G, status, (R_0, \dots, R_k))$ under tag/context $did-read$, with chain tag $did-chain$. One-byte status 0 denotes an endpoint; 1 denotes absence and an empty chain. Validate the response signature, nonce, selector, complete chain, embedded keys/expansions and endpoint. Ordered current reads select the latest endpoint; signatures alone do not establish freshness. An active endpoint returns (doc, v_D) with exact whitespace-free UTF-8 $doc = \{"id": "did"\}$ and media type `application/did+json`; all other outcomes return $\perp$. Historical resolution does not assert current control. Controller keys come from validated method records, outside the DID document [13], and are absent from anonymous presentations. Do not follow untrusted redirects.
- 5) **Issuance:** Issue. Use Section IV-B's party-specific inputs and $q_{iss} = (params, \mu, rs_e)$. The issuer validates $ev, attr$, resolves the current DID matching v_D , and obtains holder approval of the exact m^{app} , including canonical (did, v_D) , over the confidential authenticated channel. The manager permanently allocates the next four-byte $rid < 2^{20}$, advancing its counter without changing the root; exhaustion fails. Supply its zero-leaf witness and current state matching rs_e ; register a fresh instance-unused 32-byte issuer nonce n_I as pending. Define

$$\begin{aligned} Y &= H(\text{enc}_{\text{holder}}(\text{suite}, \mu, x_H)), \\ B &= (Y, E_{\text{sch}}(m^{app})), \\ M_{\text{cred}} &= \text{enc}_{\text{cred}}(\text{suite}, \mu, B, rid), \\ X_{\text{en}} &= (pp, \mu, m^{app}, rid, B, n_I, rs_e). \end{aligned}$$

B, X_{en} use binding, enrol-statement; m^{app} uses E_{sch} . The holder sends B , Prove(enrol, X_{en}, x_H) and Sign_{control}($sk_C, E(X_{\text{en}})$). The enrolment circuit checks BindOpen with a 256-bit witness; public checks enforce domains, state and equality with the approved message encoding. Verify registration, pending nonce, proof and controller signature against the resolved key. Freshly recheck the DID reference and revocation state before certification; mismatch aborts. These reads fix the issuance-time references.

Set $\sigma \leftarrow \text{Sign}_{\text{cred}}(sk_I, M_{\text{cred}})$, $\text{cert} = (B, \sigma)$ and $vc = \text{cred} = (\text{cert}, m^{app}, rid, \epsilon, \mu)$. Tags are certificate, credential; m uses E_{sch} , and

ρ is a zero-length field. BindRep validates the instance/certificate and $\rho = \epsilon$, returning B ; BindOpen checks these domains and the displayed binding equations for candidate (x_H, m) . CredValid additionally validates rid and $\text{Vfy}_{\text{cred}}(pk_I, M_{\text{cred}}, \sigma) = 1$. Bounded-verify the signature; log (μ, m^{app}, rid, B) before release, serialising logging, nonce consumption and release. The holder checks the approved vector, credential, state and witness, returning $out_H = (vc, w_e)$ with w_e stored separately. Return updated authority states even on holder failure. Abort retires the nonce, retaining allocations and released certificates; never reuse rid . The binding/certification exchange is Enroll_{DAA}; evidence/controller validation is the PQ-DID wrapper. Direct certification follows Section VI-B3; x_H stays private.

For $r < 2^{20}$, a witness $w = (s_0, \dots, s_{19})$ comprises twenty 48-byte siblings, $s_j = T_e[j, (\lfloor r/2^j \rfloor) \text{ xor } 1]$. Define PathRoot(r, b, w) = v_{20} by $v_0 = L_b$ and

$$v_{j+1} = \begin{cases} F_{j+1}(v_j, s_j), & \text{bit}_j(r) = 0, \\ F_{j+1}(s_j, v_j), & \text{bit}_j(r) = 1. \end{cases}$$

NRVerify checks all domains and PathRoot($r, 0, w$) = A_e .

- 6) **Presentation:** Present(params, vc, st_H, w_e, ctx, rs_e). Authenticate the request, obtain holder approval, and check the instance, expiry, policy, state references, credential and witness. Update outdated witnesses locally using item 8; a changed request state requires a new request. Set

$$\begin{aligned} X &= (pp, \mu, ctx, rs_e, D, m_D), \quad m_D = \text{proj}_D(m), \\ \xi &= (x_H, E_{\text{sch}}(m), rid, \sigma, w_e). \end{aligned}$$

Tag X as auth-statement; private fields occupy 32, 1024, 4, 3309 and 960 bytes. The circuit validates all private domains, reconstructs Y, B, M_{cred} , and checks

$$\begin{aligned} \text{Vfy}_{\text{cred}}(pk_I, M_{\text{cred}}, \sigma) &= 1, \\ \text{PathRoot}(rid, 0, w_e) &= A_e, \quad \text{proj}_D(m) = m_D. \end{aligned}$$

Check PubOK(X), $P_{\text{pub}}(m_D, ctx)$ publicly as in Section V-B. The same hidden rid binds certification and revocation; extraction yields $((B, \sigma), x_H, m, rid, \epsilon, w_e)$. *Bounded computation.* Execute FIPS 204 Algorithms 3 and 8 and every invoked subroutine, including context/hash processing, key expansion, signature/hint decoding, polynomial arithmetic, norm tests and challenge comparison. SHA3/SHAKE execute all FIPS 202 blocks, rounds, suffixes and padding. Caps are 1026 bytes per RejNTTPoly, 512 per RejBoundedPoly, 256 total per SampleInBall (eight sign bytes and at most 248 index bytes), and 1024 ML-DSA.Sign_{internal} iterations. Sampling preserves the standard coefficient-assignment order and consumes rejected index bytes. Exhaustion rejects verification or aborts setup/signing; bounded-verify every signature before release, without an internal retry on failure. These caps exceed the minima in FIPS 204, Appendix C [30]; accumulated

honest-operation failure remains a separate budget in Section VIII-A.

Circuit profile BC-1. The suite fixes $\text{CGen}(\text{kind}, X)$. Inline the selected relation's bounded computations in source order: evaluate operands left to right, expand matrix/vector operations componentwise, left-fold sums and traverse arrays by increasing indices. Serialised bytes enter most-significant-bit first; arithmetic bit i has weight 2^i , with FIPS bit-order conversions implemented by rewiring. Use signed 64-bit integer words, sign-extended 65-bit addition/subtraction and 128-bit products; check representability before narrowing. Reduce each ring operation modulo $q = 8380417$; preserve FIPS centred representatives where specified. Bytes and Keccak lanes remain 8- and 64-bit bitstrings. Attribute integers remain eight-byte strings inside the circuit; disclosed range checks use unsigned arithmetic outside it.

Use XOR, AND, NOT and constants, expanding $a \vee b = a \oplus b \oplus (a \wedge b)$. Ripple addition emits, low bit first, $t_i = a_i \oplus b_i$, $s_i = t_i \oplus c_i$ and $c_{i+1} = (a_i \wedge b_i) \oplus (t_i \wedge c_i)$, with $c_0 = 0$; subtraction complements the second operand and sets $c_0 = 1$. Equality left-folds bitwise XNORs; signed comparison uses the sign of the extended difference. Multiply magnitudes by summing shifted partial products in increasing bit order, then apply the sign. Division/reduction by a positive constant scans magnitude bits high to low with conditional subtraction, followed by the signed floor/remainder correction. Bit-string shifts/permutations rewire bits; integer shifts use checked multiplication/floor division by powers of two. Use $\text{mux}(s, a, b) = a \oplus (s \wedge (a \oplus b))$. Private reads scan all cells with equality selectors; writes multiplex all cells from a snapshot, preserving sequential source assignments.

Compile both branches, true before false, merging with multiplexers. Unroll array loops to their fixed capacities and sampling loops to their byte caps, retaining source initialisation, tests and increments. Share each sampler's byte counter across its invocation; termination masks later iterations. Invalid parsing, indexing or overflow sets a sticky rejection bit only on active paths; invalid reads supply zero and writes do nothing. Parallel FIPS state updates read the preceding state. Reserve wires 0 and 1 for constants, then number input bits in serialised order and gate outputs in emission order; copies reuse wires. Fold only operations with entirely public operands; retain all other gates without reassociation, common-subexpression elimination or partial constant simplification. Number AND gates in emission order; the single output is the conjunction of the required checks and absence of rejection.

The derived circuit f_X has $d = 256$ (enrol) or $d = 42632$ (auth) private bits and g AND gates. Before proving or checking, require all public domains and

$$|E(X)| \leq L, \quad V = \lceil (d + 2g)/8 \rceil \leq L.$$

The generator rejects on exceeding these bounds; no

prover-selected circuit is accepted. Thus $g \leq \lfloor (8L - d)/2 \rfloor$. Every outer-oracle input below is shorter than $B_{\max} = 2L + 2^{18}$ bytes. Admission establishes encoding capacity, not practical execution cost.

Shared prover. $\text{Prove}(\text{kind}, X, \xi)$ first derives f_X and rejects a wrong-length witness or $f_X(\xi) \neq 1$. Use 480 independent ZKBoo-style three-view repetitions [31], $j = 0, \dots, 479$, with parties $i = 0, 1, 2$ modulo three. Sample uniform ξ_0, ξ_1 and set $\xi_2 = \xi \oplus \xi_0 \oplus \xi_1$; each party receives an independent raw uniform g -bit tape. Constants share as $(a, 0, 0)$, XOR is local and NOT flips party 0. At AND gate k ,

$$z_i = (a_i \wedge b_i) \oplus (a_{i+1} \wedge b_i) \oplus (a_i \wedge b_{i+1}) \\ \oplus r_{i,k} \oplus r_{i+1,k}.$$

View $v_{j,i}$ contains its input share, tape and AND outputs in gate order: $d + 2g$ bits, packed most-significant-bit first with zero terminal padding. Reconstruct other wires locally. For output shares $y_{j,i}$, independent 64-byte salts $s_{j,i}$ and $H_{\text{RO}} = \text{SHAKE256}(\cdot, 1024)$ (128-byte output), set

$$c_{j,i} = H_{\text{RO}}(\text{enc}_{\text{view}}(\text{suite}, \text{kind}, E(X), \\ [j]_2, [i]_1, s_{j,i}, v_{j,i})).$$

A concatenates $(y_{j,0}, y_{j,1}, y_{j,2}, c_{j,0}, c_{j,1}, c_{j,2})$ in repetition order; output shares use bytes 0 or 1. After committing, sample an independent 64-byte ν and compute

$$u = H_{\text{RO}}(\text{enc}_{\text{challenge}}(\text{suite}, \text{kind}, E(X), \nu, A)), \\ e_j = \lfloor (\text{OS2IP}(u) \bmod 3^{480}) / 3^j \rfloor \bmod 3,$$

with unsigned big-endian OS2IP. Return $\pi = \nu \parallel A \parallel Z$, where Z lists the salt/view of e_j , then $e_j + 1$, for each repetition. Auth_{DAA} returns $\text{auth} = (D, m_D, \pi)$ and $vp = \text{auth}$, encoded as mask, selected fields and proof. Auxiliary relation data are $a = \eta = \epsilon$. Refresh all proof randomness, erase transient state and never reopen views under another challenge. No additional holder identifier, binding, certificate or witness is disclosed; authority/credential records remain unchanged. Failure returns $\perp$.

- 7) **Verification:** $\text{Verify}(\text{params}, vp, ctx, rs_e, st_V)$. $\text{Check}(\text{kind}, X, \pi)$ derives the admitted f_X, d, g , checks exact proof length and recomputes challenges. In each repetition, require advertised outputs to XOR to 1, both opened commitments and padding to validate, reconstructed outputs to match their advertised shares, and every AND equation of the first opened party to hold using both views/tapes. Accept only if all repetitions pass. $\text{Verify}_{\text{DAA}}(pp, vp, ctx, rs_e)$ parses (D, m_D, π) , constructs X , checks $\text{PubOK}, P_{\text{pub}}$, and invokes $\text{Check}(\text{auth}, X, \pi)$.

The DID verifier also checks exact agreement with its registered pending context, audience/session, expiry, trusted issuer/key/schema, policy and state reference. Configured DID-state checks require both did, v_D disclosed and validated resolution; hidden-DID lookups are unsupported. Finally read $\text{CurrentState}(\text{ns})$ and

match the context reference; atomically recheck the complete pending context, session and expiry, consume the challenge and return $(1, st'_V)$. Failure returns $(0, st_V)$ without this call consuming the challenge; concurrent calls may update shared state. Replicas share atomic state. The final read fixes the logical revocation epoch; subsequent updates do not invalidate past acceptance. DAA verification remains stateless.

For CurrentState, send a fresh 32-byte n_S . The manager signs $\text{enc}_{\text{current}}(\text{suite}, E(\mu), n_S, E(rs_e))$ under current for an ordered snapshot. Check nonce, instance, response signature and StateAuth; failure returns $\perp$. Request creation obtains this state, samples an audience-unused 32-byte nonce, registers ctx and signs $E(ctx)$ under request . The holder checks the pinned audience key and retains the signature outside ctx, vp . Lifetime follows application policy and trusted time. These are supporting procedures.

- 8) **Revocation:** $\text{Revoke}(\text{params}, st_R, rs_e, rid, \text{req})$. The issuer may revoke any allocated identifier, including aborted issuance. For fresh 32-byte n_R , form $\text{req} = (n_R, \beta)$ with

$$M_R = \text{enc}_{\text{revreq}}(\text{suite}, E(\mu), [rid]_4, E(\text{ref}_e), n_R), \\ \beta \leftarrow \text{Sign}_{\text{revreq}}(sk_I, M_R).$$

Check the signature, unused nonce, registration, namespace, current-state agreement and non-revocation. For $r^* = rid$, obtain its old path w^* , set its leaf to L_1 , and recompute ancestors $a_0 = L_1, \dots, a_{20} = A_{e+1}$. Sign rs_{e+1} and set $u_{e+1} = (rs_e, rs_{e+1}, r^*, w^*, v)$, tagged rupdate , where

$$M_u = \text{enc}_{\text{update}}(\text{suite}, E(\mu), E(\text{ref}_e), \\ E(\text{ref}_{e+1}), [r^*]_4, s_0^* \parallel \dots \parallel s_{19}^*), \\ v \leftarrow \text{Sign}_{\text{update}}(sk_R, M_u).$$

Validate both new signatures; atomically commit the tree, epoch, nonce consumption and public records. Return $(st'_R, rs_{e+1}, u_{e+1})$. Invalid or failed pre-commit operations return $\perp$ without state change; committed updates remain retrievable. Revocation and DID deactivation are independent.

Local witness update. Validate both state signatures, v , namespace, consecutive epochs, and $\text{PathRoot}(r^*, b, w^*) = A_{e+b}$ for $b \in \{0, 1\}$. Validate the old witness; return $\perp$ if $r = r^*$. Otherwise recompute a_j from $(r^*, 1, w^*)$ and set

$$s'_j = \begin{cases} a_j, & ([r/2^j] \text{ xor } 1 = \lfloor r^*/2^j \rfloor), \\ s_j, & \text{otherwise.} \end{cases}$$

Verify the new path. UpdateWit iterates over the ordered authenticated chain with matching endpoints; missing/inconsistent updates fail, while an empty chain preserves a valid same-state witness. Identifiers/witnesses remain local during updating; the published r^*, w^* belong to the historical privacy view.

Correctness holds for admissible executions in Section IV-C, subject to bounded-operation failures. The fixed $\lambda = 128$

profile does not itself establish overall 128-bit security. Dependencies include ML-DSA post-quantum EUF-CMA, quantum collision/preimage resistance as used, the ideal outer QROM and the stated services. Sections VIII-C and VIII-D analyse extraction [32] and joint-view privacy [33]; Section VIII-E treats lifecycle composition. Internal SHA3/SHAKE computations remain actual circuit computations.

The exact proof-length rule, used by both prover and checker, is

$$|\pi| = 64 + 480(515 + 2 \lceil (d + 2g)/8 \rceil) \text{ bytes.}$$

For authentication, the gate-independent term is 5,363,104 bytes; each million AND gates adds 240,000,000 bytes. These are calculated sizes. Generated gate counts and measured costs require circuit synthesis and execution.

VIII. SECURITY OF THE CONCRETE CONSTRUCTION

We analyse the Section VII construction in Section VI's experiments, separating certification, binding, extraction, presentation privacy and lifecycle composition. Historical privacy includes the complete subsequent revocation view.

A. Model and Quantitative Conventions

Let $H = \text{SHA3-384}$. Only the outer, domain-separated proof hash H_{RO} is modelled as a quantum-accessible random oracle with 1024-bit outputs. Internal SHA3/SHAKE computations remain the specified algorithms. Assume ML-DSA-65 post-quantum EUF-CMA security with classical signing queries and quantum collision resistance of H . Hiding a uniform x_H from its binding Y additionally requires preimage resistance on its 256-bit domain. This is separate from binding preservation. Retain Section VI-A's honest-authority, sequential-oracle, corruption and leakage restrictions, and Section IV-A's service assumptions.

Let T_A be adversarial running time, $Q \geq 1$ bound all outer-oracle queries, and N bound the proofs replaced in a privacy experiment. Q includes honest proof/checker queries, the hybrids' ordinary oracle queries and the complete post-challenge continuation. Probabilities include setup aborts, which contribute no advantage. Bounds are clipped at one; $[a]_+ = \max\{a, 0\}$.

Bounded-operation loss. Compare bounded execution $\mathcal{E}$ with the standard ML-DSA execution using the same randomness, stopping the comparison at the first exceeded cap. Budgets n_K, n_S, n_V count honest key generations, signing invocations and verifications of honestly generated signatures; n_A counts additional matrix expansions. Include pre-release verification in n_V , and all setup, controller and service operations. Reused deterministic expansions can be counted once when cached. Before the first failure, the capped subroutine counts satisfy

$$c_T \leq 30(n_K + n_S + n_V + n_A), \\ c_B \leq 11n_K, \quad c_C \leq 1024n_S + n_V, \quad c_S = n_S.$$

These follow from ML-DSA-65's 6×5 matrix, eleven short polynomials and at most 1024 challenge samplings per signing

call. Set $C = c_T + c_B + c_C + c_S$; thus $C \leq 41n_K + 1055n_S + 31n_V + 30n_A$.

Under the XOF-output and geometric-rejection model used in FIPS 204, Appendix C [30], define $F(n, p, k) = \sum_{i=0}^k \binom{n}{i} p^i (1-p)^{n-i}$. For Section VII's limits, the per-call estimates are

$$\begin{aligned}\beta_T &= F(342, 8380417/2^{23}, 255) < 2^{-594}, \\ \beta_B &= F(1024, 9/16, 255) < 2^{-304}, \\ \beta_C &= F(248, 13/16, 48) < 2^{-325}, \\ \beta_S &= (41/51)^{1024} < 2^{-322}.\end{aligned}$$

Indeed, 1026 bytes give 342 candidate field elements, and 512 bytes give 1024 candidate nibbles. After eight sign bytes, `SampleInBall` needs 49 accepted indices from 248 bytes; each acceptance probability is at least $208/256 = 13/16$. The signing estimate uses the model's conditional rejection bound $41/51$, rather than inferring a tail from the mean alone.

Let $\Delta_{\text{tail}}(\mathcal{E})$ bound the increase in the first-failure probability when these conditional tail estimates are transferred to the actual adaptive execution. The union bound gives

$$\begin{aligned}\delta_{\text{cap}}(\mathcal{E}) &\leq \Delta_{\text{tail}}(\mathcal{E}) + \sum_{j \in \{T, B, C, S\}} c_j \beta_j \\ &\leq \Delta_{\text{tail}}(\mathcal{E}) + C 2^{-304}.\end{aligned}$$

No independence between calls is required if the corresponding conditional bounds hold. In that model $\Delta_{\text{tail}} = 0$; this equality is not asserted for arbitrary adaptively selected SHAKE inputs. Malformed adversarial signatures are rejected, not charged as honest failures. Standard correctness makes pre-release failure on an honest signature a covered cap event. For a signature reduction, abbreviate

$$\varepsilon_{\Sigma}(\mathcal{B}) = \text{Adv}_{\text{ML-DSA-65}}^{\text{qEUF-CMA}}(\mathcal{B}) + \delta_{\text{cap}}(\mathcal{B}).$$

The cap term covers key generation and the relevant signing simulation; it prevents an internally signed but unreleased message from being silently classified as fresh.

Correctness, capacity and reduction cost. Use Section VII's admitted BC-1 circuits and $B_{\text{max}} = 2(2^{32} - 1) + 2^{18}$ bytes. Checked arithmetic preserves valid values: ring products are below 2^{46} and even a 256-term sum is below 2^{54} . Attribute integers remain byte strings in the circuit. For a true admitted statement, proof completeness is perfect: XORing the three AND equations cancels the tape bits and recovers the AND of the reconstructed inputs. Path-update induction gives the new root for every unrevoked credential. Operational correctness additionally requires the eligibility, state and service conditions of Section IV-C, with cryptographic cap aborts charged to δ_{cap} .

Component advantages include signing-query budgets and reduction runtime. With T_{hon} the cost of simulating the other roles and T_{gen} that of circuit generation, an extraction-based reduction takes

$$\begin{aligned}T_{\text{red}} &\leq T_{\mathcal{A}} + T_{\text{hon}} + T_{\text{gen}} \\ &\quad + O(Q^2) \text{poly}(1024, 8B_{\text{max}}) + O(480 |f_X|),\end{aligned}$$

where $|f_X|$ counts all circuit gates. Direct reductions omit the extraction terms. The factor eight converts bytes to bits.

B. Certification, Binding and Non-Revocation

Theorem 4 (Authorised certification and holder binding). *For the Section VI-B experiments, there are component reductions such that*

$$\begin{aligned}\text{Adv}_{\Pi}^{\text{cert}}(\mathcal{A}) &\leq \varepsilon_{\Sigma}(\mathcal{B}_{\text{cert}}), \\ \text{Adv}_{\Pi}^{\text{bind}}(\mathcal{A}) &\leq \text{Adv}_H^{\text{qCR}}(\mathcal{B}_{\text{bind}}).\end{aligned}$$

Proof. Embed the challenge key as pk_I , forward its signing queries with their contexts, and simulate the other roles. Compare this standard-oracle simulation with bounded signing, losing at most δ_{cap} ; the simulation aborts on a failed pre-release check. No access to the oracle's secret internal counters is assumed.

Outside the coupling failure, every released credential signs $M = \text{enc}_{\text{cred}}(\text{suite}, \mu, B, \text{rid})$ with $B = (Y, E_{\text{sch}}(m^{\text{app}}))$, and the issuer logs $(\mu, m^{\text{app}}, \text{rid}, B)$ before release. If a winning opening's message was already signed, injective encodings and $\text{BindOpen} = 1$ make it match that retained record, contradicting Authorised = 0. Otherwise it gives a fresh credential-context EUF-CMA forgery; other contexts cannot supply it. This directly certifies the approved vector without extracting every enrolment witness. External evidence validation remains trusted.

Two valid openings of one B have identical canonical attributes and $\rho = \epsilon$. A binding win therefore has $x_0 \neq x_1$ but equal hashes of their distinct $\text{enc}_{\text{holder}}(\text{suite}, \mu, x_i)$ inputs. Output this collision. Voluntary sharing of the original opening is outside that experiment. $\square$

Theorem 5 (Sound non-revocation paths). *In Section VI-D's authenticated history,*

$$\text{Adv}_{\Pi}^{\text{nm}}(\mathcal{A}) \leq \text{Adv}_H^{\text{qCR}}(\mathcal{B}_{\text{tree}}).$$

Proof. Maintain the actual tree at every epoch. A winning (e, r, w) reconstructs A_e from L_0 , although $r \in \mathcal{R}_e$ and its genuine path starts at L_1 . If the leaves coincide, their distinct encodings collide. Otherwise the first level at which the two running hashes coincide gives distinct ordered child inputs with an equal level-tagged hash. This works for the adversarially chosen epoch and all permitted updates and corrupted-holder paths. $\square$

C. Complete-Witness Knowledge in the QROM

Theorem 6 (Extraction preserving the target event and history). *For Section VI-B4's target predicate S and success probability ϵ_S , set*

$$\begin{aligned}p_{\star} &= (2/3)^{480} + 2^{-544}, \\ \varepsilon_{\text{ex}}(Q) &= \min\{1, 31740Q^3 2^{-1024} + 20Q^2 p_{\star}\}, \\ F_{\text{auth}}(\epsilon_S; 128, Q) &= [\epsilon_S - \varepsilon_{\text{ex}}(Q)]_+.\end{aligned}$$

The profile satisfies that extraction contract with knowledge error $\kappa(Q) = \varepsilon_{\text{ex}}(Q)$ and the running time above.

Proof. Fix the statement, 1440 committed messages and advertised outputs. The canonical commitment inputs bind the suite, proof kind, statement, repetition, party, salt and view; the outputs and nonce are additional first-message data. If all three pair checks accept in one repetition, all three parties' AND

equations hold. Gatewise XOR reconstructs ξ with $f_X(\xi) = 1$. Testing three pairs in each repetition finds such a witness in polynomial time.

This verifies $\mathcal{S}$ -soundness* in [32, Definition 3.5], where a challenge set belongs to $\mathcal{S}$ if some repetition's projection contains all three choices. Otherwise at most 2^{480} challenge vectors accept. Each vector has probability at most $3^{-480} + 2^{-1024}$ under the specified modulo map, giving p_* . The ordinary commit-and-open theorem [32, Theorem 4.2] therefore gives the displayed error, since $22 \cdot 1440 + 60 = 31740$.

For the oracle environment, execute the adversary and real honest algorithms together, using actual witnesses for honest proofs. Count their queries in Q , retain the classical authorisation, revocation and presentation histories, and forward external classical component-oracle calls without rewinding. Apply the theorem to the adaptively selected terminal statement, retaining those histories as auxiliary output. Its compressed-oracle simulation has zero simulation error, and acceptance with invalid extraction has probability at most ε_{ex} . Restricting this event to fresh statements satisfying $\mathcal{S}$ gives the claimed lower bound without changing the target history.

Decode $\xi = (x_H, E_{\text{sch}}(m), \text{rid}, \sigma, w_e)$, reconstruct Y, B , and output

$$\omega_e = ((B, \sigma), x_H, m, \text{rid}, \epsilon, w_e).$$

The circuit enforces domains, signature validity, disclosures and non-revocation for the same certified rid ; public verification enforces $\text{PubOK}, P_{\text{pub}}$. Thus $\mathcal{R}_{\text{auth}}(X, \omega_e) = 1$. Extraction uses the execution and simulated oracle, rather than a transcript alone. The same reconstruction applies to enrolment; certification does not require simultaneous issuance extraction. $\square$

D. Adaptive Simulation of Presentations

For the 512-bit salts and proof nonce, define

$$\alpha(Q) = \sqrt{Q 2^{-512}} + \frac{1}{2}Q 2^{-512},$$

$$\delta_{\text{ZK}}(N, Q) = \min\{1, N(481\alpha(Q) + 960Q 2^{-256})\}.$$

Theorem 7 (Online simulation of the complete proof view). *At most N adaptively requested proofs of true admitted statements can be simulated online from their public statements and proof kinds, with distinguishing loss at most $\delta_{\text{ZK}}(N, Q)$ over the entire permitted continuation. No witness, hidden candidate index or future revocation data are required.*

Proof. Retain real persistent records throughout these games.

G_0 : *real proofs.* Run the prescribed prover and subsequent oracle operations.

G_1 : *programmed challenges.* After committing the views, sample uniform $u \in \{0, 1\}^{1024}$, derive the specified modulo challenges, and program their hash point using the fresh 512-bit ν . Its conditional maximum point probability is 2^{-512} , so adaptive reprogramming [33, Theorem 1] gives $|\Pr[G_0 = 1] - \Pr[G_1 = 1]| \leq N\alpha(Q)$, including subsequent oracle access. Since view generation does not use u , it can equivalently be sampled first.

G_2 : *uniform unopened commitments.* Replace each repetition's unopened commitment by an independent 1024-bit

string. To justify one replacement, first program that value when its independent salt is sampled, costing at most $\alpha(Q)$. Then remove this programming while retaining the random commitment. In the unprogrammed comparison no operation uses the hidden salt; even given later credential/view information, its query-point probability remains at most 2^{-512} . A telescoping quantum-query hybrid bounds the change by $2Q 2^{-256}$. Consequently,

$$|\Pr[G_1 = 1] - \Pr[G_2 = 1]| \leq 480N(\alpha(Q) + 2Q 2^{-256}).$$

This uses erasure and the prohibition on reopening views; challenge programming is in a separate domain.

G_3 : *witness-free views.* For each selected pair, sample input shares and raw tapes uniformly; sample the second party's AND outputs uniformly and compute the first party's outputs from its checked equations. Reconstruct the other gates and choose the third output share to make XOR one. The missing input share is one-time masked, and every unchecked AND output has a fresh unknown tape bit. This is exactly the valid two-view distribution [31, Proposition 4.1]. Hash the opened views with fresh salts, choose unopened commitments uniformly, form A , sample ν , and program the challenge point to the previously sampled u . Sampling u preserves the exact modulo distribution; canonical packing preserves proof lengths. Thus G_3 equals G_2 and uses only public inputs. Adding the losses proves the claim, including the residual quantum state. $\square$

Theorem 8 (Adaptive presentation privacy). *In Section VI-C's experiment,*

$$\text{Adv}_{\Pi}^{\text{priv}}(\mathcal{A}) \leq \delta_{\text{ZK}}(N, Q).$$

This includes its anonymity, unlinkability and selective-disclosure specialisations.

Proof. In each world b , simulate the left-right responses using Theorem 7. Admissibility supplies identical public statements and outcomes; eligibility and witness maintenance check both candidates before selection. Presentation changes no persistent credential/authority record. Every remaining named oracle therefore runs without b , including under verifier collusion, changing supported policies and background revocations. Induction makes the simulated worlds identical. Section VI's guessing advantage is half their real-world probability difference, so averaging the two simulation losses gives the bound. $\square$

Theorem 9 (Post-revocation unlinkability). *For Section VI-D2's two-phase experiment,*

$$\text{Adv}_{\Pi}^{\text{pru}}(\mathcal{A}) \leq \delta_{\text{ZK}}(N, Q),$$

where N counts historical challenge proofs and Q includes the entire post-cut-off continuation.

Proof. Simulate historical challenge proofs while retaining both actual candidate records. After the cut-off, execute real named revocations, publishing the signed states, revoked identifiers and old sibling paths, together with all permitted updates, named presentations and background corruptions. These operations may depend on earlier observations but never

receive the historical selection bit. Hence the joint simulated history, subsequent revocation publications and auxiliary information are independent of b . Theorem 7 bounds each complete real-world distance, supplying the two terms of Theorem 1. Their average gives the result. Protected-holder corruption and hidden-index presentations after the cut-off remain forbidden; no claim of post-compromise privacy follows. $\square$

E. Composition with the Concrete DID Lifecycle

Let ε_{sys} bound cryptographic authenticity failure in the signed-service realisation, retaining honest ordering, latest-state, time and atomic-consumption services. A conservative multi-key bound is

$$\varepsilon_{\text{sys}} \leq \sum_{k \in \mathcal{K}_{\text{sys}}} \varepsilon_{\Sigma}(\mathcal{B}_k) + \text{Adv}_H^{\text{qCR}}(\mathcal{B}_{\text{record}}) + q_n(q_n - 1)2^{-257}.$$

$\mathcal{K}_{\text{sys}}$ comprises relevant honest issuer, revocation, registry, controller and request keys; each reduction includes every signing purpose under its key and aborts if that key is corrupted before the target violation. The nonce term covers at most q_n independent 256-bit read nonces; retained used-set checks exclude successful repeats for other nonce types.

At the first accepted unauthorised service message, its context and canonical contents give a fresh signature forgery, unless distinct hash-referenced records share a digest, giving a collision. Genesis, successor, endpoint, selector and nonce checks exclude other substitutions and replays outside these events. Signatures do not establish ordering or freshness. In Section VI's ideal authentic-service experiments, $\varepsilon_{\text{sys}} = 0$.

Theorem 10 (Lifecycle authentication and revocation soundness). *For target acceptance probability ϵ on a fresh base statement at an honest verifier, compatible reductions give*

$$p_{\text{good}} \geq [\epsilon - \varepsilon_{\text{sys}} - \varepsilon_{\text{ex}}(Q) - \varepsilon_{\Sigma}(\mathcal{B}_{\text{cert}}) - \text{Adv}_H^{\text{qCR}}(\mathcal{B}_{\text{tree}})]_+,$$

where p_{good} is the probability of extracting an authorised complete witness unrevoked at the final state read. The bound also holds for the permitted efficient terminal predicates of Section VI-B4.

Proof. Outside system-authenticity failure, acceptance uses the expected context, trusted instance and root at the final current-state read. Theorem 6 preserves this event and history. An extracted opening absent from the authorisation table wins Theorem 4; an extracted certified rid revoked at that read wins Theorem 5. Subtracting these events instantiates Theorem 2.

The approved vector retains the validated issuance-time (did, v_D) binding, without asserting later control of a hidden DID. Atomic challenge consumption prevents repeated acceptance; fresh contexts require the corresponding statement-bound proof. Voluntary sharing and live relay remain outside this knowledge claim. $\square$

Theorem 11 (Privacy under the DID lifecycle). *For $g \in \{\text{priv}, \text{pru}\}$ under the corresponding Section VI restrictions,*

$$\text{Adv}_{\text{PQ-DID}}^g(\mathcal{A}) \leq \delta_{\text{ZK}}(N, Q) + \varepsilon_{\text{sys}}.$$

Proof. Couple the signed-service execution to authentic services, then simulate its challenge proofs. Maintain actual DID histories, registry/controller records, current-state replies, pending contexts and revocation publications. Named services require no hidden candidate index. A hidden DID triggers no lookup; any disclosed (did, v_D) and resolution outcome must belong to the common authorised leakage. Thus the authentic wrapper contributes $\Delta_{\text{wrap}, b} = 0$ in Theorem 3. Apply the corresponding presentation or historical-privacy result and add the coupling loss. The model's excluded authority collusion, holder compromise and network correlation remain excluded. $\square$

Illustrative resource evaluation. Consider ceilings $T_A \leq 2^{80}$, $N \leq 2^{32}$, $|\mathcal{K}_{\text{sys}}| + 1 \leq 2^{20}$ and $q_n \leq 2^{32}$. Let the sum of the capped-call budgets across all cap terms in the chosen bound be at most 2^{64} . This counts their reduction simulations, rather than multiplying one execution's budget silently. Conservative upper bounds for the ideal-oracle proof terms are

Q	$\varepsilon_{\text{ex}}(Q)$	$\delta_{\text{ZK}}(2^{32}, Q)$
2^{64}	$< 2^{-148}$	$< 2^{-150}$
2^{80}	$< 2^{-116}$	$< 2^{-134}$

The aggregate cap term is at most $2^{-240} + \sum \Delta_{\text{tail}}$, and the read-nonce term is below 2^{-193} . The table evaluates ideal-oracle bounds. An overall security claim must additionally justify the component advantages at reduction runtime, the sampling-model discrepancy and the service assumptions.

IX. IMPLEMENTATION AND EVALUATION

A. Prototype Implementation

To evaluate the feasibility of PQ-DID, we implemented a proof-of-concept prototype in Python. The prototype realises ML-DSA-65 credential certification, lattice commitments and re-randomisation, a witness-dependent Fiat-Shamir proof structure, selective disclosure, presentation verification, and revocation checking. It employs the PQ-Commit-ZKP framework based on BDLOP commitments and the LNP sigma protocol, SHAKE-based hashing for Fiat-Shamir, and a revocation component for credential-status checking. The current prototype does not natively encode ML-DSA verification or accumulator non-membership verification inside the lattice proof; these predicates are captured by the Expressive PQ-ZK Relation Support assumption in Section VIII-A.

All experiments were conducted on an Intel Core i9-14900HX processor with 16 GB RAM running Windows 11.

B. Experimental Results

We conducted 100 independent trials for each protocol phase. Table II reports the average execution times.

All authentication attempts generated during the experiments were successfully verified. The implementation also confirmed selective disclosure and revocation checking: credentials accepted before revocation were rejected after revocation.

TABLE II
AVERAGE EXECUTION TIME OVER 100 TRIALS.

Operation	Time (s)
Setup	0.003533
Enrollment	0.000900
Authentication	0.000130
Verification	0.000331

TABLE III
SIZE OF CRYPTOGRAPHIC OBJECTS.

Object	Size (bytes)
Issuer public key	1952
Issuer secret key	4032
ML-DSA signature	3309
Credential	4157
Authentication token	4677

C. Communication and Storage Costs

Table III summarises the storage and communication costs of the implementation.

The authentication-token overhead is dominated by the privacy-preserving proof and its associated post-quantum proof data, the issuer signature remains hidden within the authentication witness.

D. Practical Interpretation

The prototype shows that verification remains below one millisecond in the tested setting, while the main overhead comes from post-quantum credential certification and zero-knowledge proof generation. The authentication token is larger than a classical DID challenge-response message because it contains post-quantum cryptographic material and a privacy-preserving proof. This overhead is expected in exchange for post-quantum security, selective disclosure, unlinkability, and revocation support.

These results indicate that PQ-DID is suitable for high-assurance identity applications, such as digital government credentials, cross-border identity verification, financial compliance, and enterprise identity systems, where privacy and long-term security outweigh the larger post-quantum communication cost.

E. Comparison and Complexity Analysis

We compare the proposed framework with representative decentralised identity systems, including W3C DID/VC 2.0 with Bitstring Status List v1.0, Hyperledger Indy AnonCreds, Transition of Self-Sovereign Identity to Post-Quantum Cryptography, and Post-Quantum Blockchain-Based DID for ORAN. The comparison considers both functionality and complexity characteristics.

The entries labelled “Not considered” indicate that the corresponding property is not explicitly defined, analysed, or formally evaluated in the referenced work. The entries labelled “Not reported” indicate that the corresponding work does not provide explicit asymptotic or concrete measurements for that metric.

Table IV shows that existing decentralised identity systems typically focus either on practical deployment or post-quantum

migration. W3C DID and Hyperledger Indy provide mature deployment frameworks but rely on classical cryptographic assumptions. Existing post-quantum DID proposals primarily focus on replacing classical cryptographic primitives with post-quantum alternatives, but generally do not provide formal security models, anonymous authentication, unlinkability, or selective disclosure mechanisms.

In contrast, the proposed framework integrates a dedicated post-quantum decentralised anonymous authentication layer into a formally defined PQ-DID architecture. The resulting construction simultaneously provides post-quantum security, anonymous authentication, unlinkability, selective disclosure, revocation support, formal security definitions, provable security reductions, a concrete cryptographic construction, and a prototype implementation.

Credential issuance requires one issuer-signature operation. Each presentation performs commitment re-randomisation, proof generation, and non-revocation processing, while verification checks the proof against the current accumulator state. For Table IV, complexity is measured with respect to the number N of issued credentials, for fixed security and credential parameters; hence these costs are $O(1)$ in N .

X. DISCUSSION AND FUTURE WORK

A. Discussion

PQ-DID differs from conventional DID authentication by separating identity management from authentication. In many DID deployments, a holder proves control of a public verification key recorded in a DID document, which can introduce structural linkability when the same key or identifier is reused across verifiers.

PQ-DID instead adopts a credential-centric authentication model. DID documents remain responsible for identifier resolution and public metadata, while authentication is delegated to PQ-DAA and performed through issuer-certified credentials, re-randomisable commitments, and zero-knowledge proofs. This design enables holders to prove credential validity and selectively disclose attributes without exposing persistent authentication identifiers.

The main trade-off is communication size: post-quantum privacy-preserving authentication is larger than conventional signature-based DID authentication. Nevertheless, the prototype results show that the overhead remains practical for high-assurance decentralised identity applications requiring post-quantum security, unlinkability, selective disclosure, and revocation support.

B. Future Work

Several directions remain open for future research.

a) *Efficiency Optimisation*: Future work may investigate parameter optimisation, proof aggregation techniques, and implementation improvements to further reduce computational and communication overhead.

b) *Advanced Post-Quantum Zero-Knowledge Proofs*: Exploring more compact lattice-based proof systems and recursive proof constructions may improve scalability while preserving security guarantees.

TABLE IV
COMPARISON WITH REPRESENTATIVE DECENTRALISED IDENTITY SYSTEMS.

Scheme	Post-Quantum	Anonymous Authentication	Unlinkability	Selective Disclosure	Revocation	Formal Security Model	Concrete Construction	Credential Size	Authentication Size	Verification Cost
W3C DID/VC 2.0 + Bitstring Status List v1.0	No	Not considered	Not considered	Not considered	Yes	No	No	Not reported	Not reported	Not reported
Hyperledger Indy Anon-Creds	No	Yes	Yes	Yes	Yes	Not considered	Yes	$O(1)$	$O(1)$	$O(1)$
Transition of SSI to PQC	Yes	Not considered	Not considered	Not considered	Not considered	No	No	Not reported	Not reported	Not reported
PQ Blockchain-Based DID for O-RAN	Yes	Not considered	Not considered	Not considered	Not considered	No	Yes	Not reported	Not reported	Not reported
This work	Yes	Yes	Yes	Yes	Yes	Yes	Yes	$O(1)$	$O(1)$	$O(1)$

c) *Scalable Revocation Mechanisms*: Future work may investigate concrete post-quantum accumulator constructions supporting efficient witness generation, witness updates, and large-scale credential revocation management.

d) *Standard-Model Security and Extended Functionality*: Constructing comparable privacy-preserving decentralised identity mechanisms with security in the standard model remains an important open problem. Future extensions may also support richer credential predicates, credential aggregation, threshold issuance, and multi-issuer credential ecosystems.

XI. CONCLUSION

This paper presented PQ-DID, a generic framework for post-quantum decentralised identity, together with PQ-DAA, a privacy-preserving authentication layer for anonymous credential presentation. The proposed architecture addresses a fundamental gap between existing decentralised identity systems, which remain largely based on classical cryptography, and existing post-quantum authentication schemes, which do not capture the complete credential lifecycle required by DID ecosystems.

By separating identity management from credential-based authentication, DID documents are no longer used as authentication credentials but instead function solely as identifier-resolution metadata, while holder authentication is realised through issuer-certified credentials. PQ-DID supports W3C-compatible workflows while enabling selective disclosure, unlinkable multi-verifier authentication, revocation checking, and forward privacy under post-quantum assumptions. We formalised the syntax and security requirements of PQ-DID and PQ-DAA, provided game-based security analyses in the QROM, and instantiated the framework using lattice-based cryptographic primitives including ML-DSA, lattice commitments, zero-knowledge proofs, and dynamic accumulators.

The prototype implementation demonstrates that privacy-preserving decentralised identity can be realised with practical post-quantum primitives. Overall, PQ-DID provides a foundation for building decentralised identity infrastructures that remain secure and privacy-preserving in the post-quantum era.

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XII. BIOGRAPHY SECTION



Yujing Sun received the B.IT. degree (First Class Honours) and the Ph.D. degree in Information Technology from Multimedia University (MMU), Malaysia, in 2020 and 2025, respectively. She is currently a Research Fellow with the Digital Trust Centre, Nanyang Technological University (NTU), Singapore.

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Yujing Sun is a Senior Research Fellow at Digital Trust Centre (DTC, Singapore National AI Safety Institute), Nanyang Technological University. She oversees DTC's technical initiatives and works closely with Prof. Kwok Yan Lam (Associate Vice President of NTU and Executive Director of DTC). Prior to joining NTU, she was a Research Assistant Professor at Department of Computer Science, the University of Hong Kong, working closely with Prof. Siu Ming Yiu. She has published more than 30 papers on top-tier venues including TPAMI, NIPS, ICLR, CVPR, ECCV, TVCG, TIP, etc. Dr. Sun received her B.S. from the University of Minnesota Twin Cities in the U.S and her Ph.D. in Computer Science from the University of Hong Kong under the supervision of Prof. Wenping Wang (IEEE & ACM Fellow).

Her research interests span Artificial Intelligence, AI Safety, Privacy-Enhancing Technologies, and Financial and Blockchain Anti-Crime Technologies.



Poh-Wen Kho received the B.IT. (Hons.) degree in Security Technology from Multimedia University, Malaysia, in 2023. He is currently pursuing the Ph.D. degree in Information Technology at Multimedia University, where he also serves as a Graduate Research Assistant.

His doctoral research focuses on the design, formal security analysis, implementation, and evaluation of secure data-sharing protocols for cyber-physical cloud environments. His research interests include applied cryptography, identity-based cryptography, searchable encryption, cloud security, provable security, and post-quantum cryptography.



Huaxiong Wang is the President's Chair in Cryptology and a Professor in the School of Physical and Mathematical Sciences at Nanyang Technological University (NTU), Singapore, where he has been a faculty member since 2006. He received a Ph.D. in Mathematics from the University of Haifa, Israel, in 1996, and a Ph.D. in Computer Science from the University of Wollongong, Australia, in 2001. He is the Director of the Strategic Centre for Research in Privacy-Preserving Technologies and Systems (SCRIPTS) and Co-Director of Singapore's

National Centre for Research in Digital Trust. Before joining NTU, he held faculty positions at Macquarie University and the University of Wollongong in Australia. Professor Wang's research interests include cryptography, cybersecurity, post-quantum cryptography, fully homomorphic encryption, secure multiparty computation, and privacy-preserving artificial intelligence. He served as Program Co-Chair of Asiacrypt 2020 and Asiacrypt 2021 and was elected a Fellow of the International Association for Cryptologic Research (IACR) in 2026.



Kwok-Yan Lam (Senior Member, IEEE) is currently the Associate Vice President (Strategy and Partnerships) and full professor in the College of Computing and Data Science with the Nanyang Technological University, Singapore. He received the BSc degree (1st Class Hons.) from the University of London, in 1987, and the PhD degree from the University of Cambridge, in 1990. He was a visiting scientist with the Isaac Newton Institute, Cambridge University, and a visiting professor with the European Institute for Systems Security. He is

also the executive director of the Digital Trust Centre Singapore (DTC), executive director of the Singapore AI Safety Institute, and director of the Strategic Centre for Research in Privacy-Preserving Technologies and Systems (SCRiPTS). He has been a professor of the Tsinghua University, PR China (2002–2010) and a faculty member of the National University of Singapore and the University of London since 1990. From 2020, he is on part-time secondment to the INTERPOL as a consultant with Cyber and New Technology Innovation. His research interests include distributed systems, intelligent systems, IoT security, distributed protocols for blockchain, homeland security, cybersecurity, AI safety. In 1998, he was the recipient of the Singapore Foundation Award from the Japanese Chamber of Commerce and Industry in recognition of his research and development achievement in information security in Singapore and the 2022 Singapore Cybersecurity Hall of Fame Award.